

Green Synergy: A Three-Party Fashion Retail System Balancing Profit Maximization and Environmental Impact Minimization

(Authors' names are not included for peer review)

Abstract. Problem definition: The fashion industry must drastically reduce its environmental impact. Yet, confusion and misunderstanding among consumers, retailers, and policymakers about key dimensions of environmental impact hinder progress. In this context, we study how to design and price a product line that balances profitability and environmental impact.

Methodology/results: We measure preferences and willingness-to-pay (WTP) for three environmental attributes—Recycled Content, Durability, and Circularity Design—through an incentive-aligned choice-based conjoint experiment and Hierarchical Bayes modelling. We characterize the trade-off between profitability and environmental impact improvements through a counterfactual simulation and a joint product line selection and pricing optimization. Our experimental design also features a between-subjects manipulation, showing half of the participants a short educational video, to assess the impact of increased consumer awareness. While price emerges as the primary driver of choice, we find that environmental attributes, though not well understood, are of clear interest and are associated with a meaningful willingness to pay. **Managerial implications:** Our findings suggest that retailers can already achieve modest reductions in environmental impact without sacrificing profitability. Furthermore, our counterfactual simulation reveals that increasing consumer awareness flattens the trade-off between profitability and environmental performance, while also encouraging more selective purchasing behaviour. For both retailers and policymakers, our study provides a deeper understanding of consumer preferences and WTP for environmental attributes, and establishes a clear hierarchy of their importance in the fast fashion context.

Key words: sustainability, fashion retail, operations-marketing interface, systems approach, data-driven optimization

1. Motivation

There have been rising calls for the fashion industry to lower its environmental impact, driven by growing pressure from consumer groups that demand more transparency, a transition to less impactful processes and materials, and a shift from the current linear *take-make-waste* approach to a circular economy model. In response, the industry has tried to increase the availability of “greener” or “more sustainable” products to meet the evolving demand from heightened consumer scrutiny. However, an important hurdle in achieving any real progress lies in the concept of environmental impact within the fashion industry: a concept intricate, multi-dimensional, and vague in its understanding to consumers, retailers, and policymakers.

Consumers appear to be well aware of the environmental challenges faced by the fashion industry. However, fashion retailers often lack sufficient incentive to integrate sustainable practices into their operations.

The key obstacle lies in the gap between consumers' stated intentions and their actual purchasing behaviour. While surveys report that sustainability is a top priority for many consumers—and that they express a willingness to pay more for “sustainable” or “eco-friendly” fashion items ([Global Fashion Agenda and BCG 2019](#), [Statista 2021](#))—retailers argue that, in practice, consumers are reluctant to pay a premium for products with lower environmental impact. According to [Global Fashion Agenda and BCG \(2017\)](#), low willingness to pay is cited as the primary barrier to adopting sustainable production, with lack of consumer awareness ranked fourth. Additionally, the absence of effective regulations is identified as the second major barrier. As a result, the current market dynamics provide fashion retailers with limited motivation to make meaningful sustainability improvements.

While there have been efforts from retailers to increase consumer awareness through sustainability icons or claims, regulators are cracking down on some over concerns of greenwashing.¹ Yet, governments and policymakers have only just begun to develop strategies and regulations to curtail the fashion industry's practices and impact. For example, the European Union adopted in March 2022 the *EU Strategy for Sustainable and Circular Textiles* ([European Commission 2022](#)). However, the Strategy remains vague on the specifics as to enactment of regulations, and how targets will be achieved and enforced.

Adding to the confusion, there seems to be no consensus as to which dimension of environmental impact (e.g., sourcing, durability, circular economy) is most important. This may be partly due to the multi-dimensional, multi-level nature of environmental impact that encompasses different stages of a garment's lifecycle, making it difficult to prioritise policies and actions. Policymakers, through proposed regulations, and consumer groups in their calls to action, seem to regard all types of environmental impact as equally important. It is unclear, however, if consumers perceive these dimensions as equally important, or if there is a hierarchy in preferences and salience. Also, it is unclear how financially feasible it is for retailers to offer all dimensions simultaneously, if required.

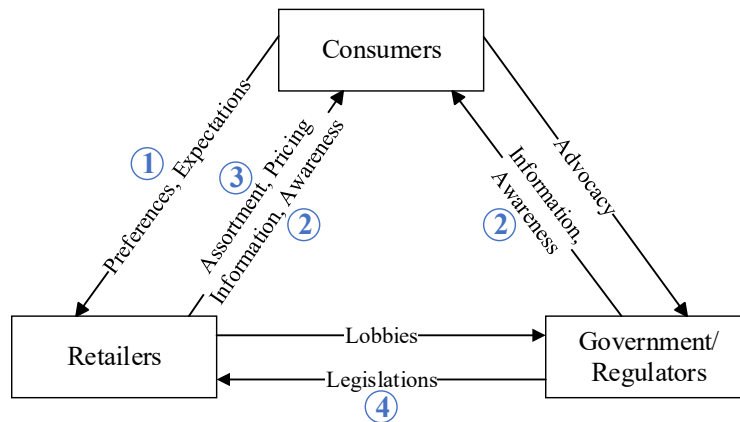
As a result, consumers have difficulty in understanding and appraising environmental attributes, retailers struggle to decide what to offer in view of uncertain consumer preferences and WTP for environmental attributes and the often-costly investments required to offer them, and policymakers struggle defining regulations that can usher both consumers and retailers into more environmental-friendly behaviour.

In this paper, we adopt a system-focused approach incorporating the consumer, retailer, and policymaker perspectives to study behavioural and environmental impacts of current fashion consumption and product line decision, and evaluate how policy changes might affect the dynamics among the three parties involved. Figure 1 illustrates this three-party system. Our experimental, data-driven optimization approach focuses

¹For example, the [European Commission](#) asked [Zalando](#) to remove sustainability labels over concerns they were misleading. In 2022, the [Norwegian Consumer Authority](#) asked [H&M](#) to remove sustainability claims deemed misleading.

on three main research questions: (RQ1) *What are consumer preferences and WTP for environmental attributes?*, (RQ2) *How is a retailer's profitability impacted by restrictions placed on these environmental attributes?*, and (RQ3) *To what extent can increased consumer awareness on environmental attributes soften the trade-off between profitability and environmental impact performance for retailers?*

Figure 1 Three-Party Fashion Retail System



Note. The numbered arrows correspond to the dimensions of Table 1 contrasting this study to prior work.

We measure consumer preferences for reduced environmental impacts in fashion products through an incentive-aligned, choice-based conjoint (IA-CBC) experiment that focuses on three attributes having the potential to reduce environmental impacts (referred to as *environmental attributes*): Recycled Content, Durability, and Circularity Design. Our experiment leverages a between-subjects manipulation, where approximately half of the respondents receive an experimental treatment: a brief educational video explaining how environmental attributes can reduce the environmental impacts of fashion garments. We use a Multinomial Logit model estimated with Hierarchical Bayes (HB-MNL) to infer individual-level consumer preferences and WTP for environmental attributes, leveraging this model to assess the treatment impact on preferences and WTP. Building on the estimated consumer preferences and current regulations, we develop the retailer's joint product line selection and pricing optimisation problem subject to constraints on environmental improvements. Our proposed formulation allows efficient computation using commercial solvers while incorporating heterogeneity in consumer preferences. By progressively increasing the minimal threshold on improvements, we reveal the profitability-environmental impact trade-off frontier. Using a counterfactual simulation, we then compare this trade-off across two scenarios: one where consumer awareness is increased through our experimental manipulation, and one reflecting the status quo.

Our results provide a rich characterization of preferences for (different levels of) environmental vis-à-vis base attributes (e.g., Price and Bill of Materials) and consumer heterogeneity. We find that although

environmental attributes are not the primary drivers in consumer choice and that they are still misunderstood, there is interest and a willingness to pay for them. Our experimental results show Recycled Content to have the highest utility, followed by Durability and Circularity Design, contradicting what respondents report by ranking Durability (then Recycled Content) as the most relevant environmental attribute. Furthermore, we find that for all environmental attributes, partworth utilities (and willingness to pay) follow a step-function pattern, with a significant initial increase in value, but diminishing returns at higher levels. Finally, our data reveal four distinct consumer segments with contrasting preferences for both base and environmental attributes. Notably, approximately 40% of respondents fall into an environmentally-leaning segment. This group tends to include more women and is generally characterised by lower education and socioeconomic status (SES). These environmentally-conscious consumers are also frequent fashion shoppers.

We complement these behavioural insights by analysing various combinations of threshold constraints on environmental improvements within a joint product line and pricing optimization. Our results reveal that retailers can already offer modest levels of environmental attributes without sacrificing profitability, despite generally low consumer sensitivity to such features. Our counterfactual simulation further shows that exposing respondents to a short educational video (the treatment) mitigates the trade-off between profitability and environmental impact. It also makes consumers more selective, increasing the likelihood that they forgo purchases of high-impact products. These findings highlight an opportunity for collaboration among consumer advocacy groups, retailers, and regulators to co-create policies that support a sustainable transition.

Overall, our study provides a robust strategic economic case for sustainability in the fashion industry, echoing recent calls from thought-leaders (Cool et al. 2024). Moreover, our integrated three-party systems approach—generalizable to other industries—offers a practical decision-support tool that incorporates behavioural insights to help retailers balance environmental improvements with profitability goals by accounting for the interplay between consumers, retailers, and policymakers.

The remainder of the paper is organized as follows. In §2, we review the relevant literature pertaining to our study while §3 introduces our integrated framework and experimental methodology. Next, §4 presents the main experimental results of the study, and §5 presents our optimization model, the counterfactual simulation, its results, as well as two robustness checks. Finally, §6 discusses the results, summarizes managerial insights and policy implications, and concludes with limitations and areas of future research.

2. Relevant Literature

Our study contributes to the Marketing literature on consumer preferences for sustainable attributes, as well as to the growing Operations Management (OM) literature on sustainable operations. As summarised in

Table 1, sustainable fashion remains an emerging topic with limited academic attention. By addressing the interactions among consumers, retailers, and policymakers, our study highlights the importance of aligning market preferences, business incentives, and regulatory frameworks to advance sustainability in the fashion industry.

Table 1 Relevant Literature on Consumer Preferences and WTP for Sustainability and Sustainable Operations in Fashion

Study	Apparel type	Consumer perspective (preference measurement) ¹					Retailer perspective		Regulatory perspective ⁴	
		Methodology	Preference heterogeneity	WTP calculation	Incentive-alignment	Manipulation ²	Decision ³	Methodology		
Measuring consumer preferences and WTP for sustainability in fashion										
Brand and Rausch (2021)	Outdoor apparel	CBC exp.	x	x						
Rausch and Kopplin (2021)	Apparel (undefined)	MaxDiff								
Wang et al. (2022)	Outdoor apparel	CBC exp.	x	x						
McKie et al. (2023)	Fast fashion	Survey				Written information, between-subjects				
Sustainable operations in fashion										
Long and Nasiry (2022)	Fast fashion		x					Product quality, variety, inventory	Analytical newsvendor model	Disposal cost, production tax
Long and Gui (2024)	Fast fashion		x					Fabric purchase, inventory	Analytical newsvendor model	Subsidies, Destruction ban
Alptekinoğlu and Örsdemir (2022)	Apparel		x				Product variety, pricing, inventory	Analytical model	Disposal cost, mandatory recycling	
Alptekinoğlu et al. (2023)	Fast fashion	Survey								
This study	Fast fashion	CBC exp.	X	X	X	Awareness video, between-subjects	Assortment, pricing	Data-driven optimization	Environmental improvement constraints	

Note. Superscript numbers refer to designated arrows of Figure 1

Measuring Consumer Preferences and WTP for Sustainability in Fashion. Motivated by the heightened consumer scrutiny and demand for sustainable features in garments, several recent studies have been concerned with understanding “green” or “eco-friendly” consumption behaviour within fashion (Kang et al. 2013, Rausch and Kopplin 2021). However, few studies have investigated specific preferences for sustainable attributes in garments.

Measuring Preferences for Sustainability. Most of the studies looked at outdoor apparel, and reveal no consensus on the environmental attribute most important for consumers. Brand and Rausch (2021) look at sustainability attributes in outdoor apparel to find materials is the most important sustainable attribute. Rausch et al. (2021) find that, when weighing sustainable versus conventional attributes in apparel, consumers place greater importance on conventional attributes. Among sustainable attributes, they find durability is most preferred. It remains unclear, however, which type of apparel (e.g., athletic vs. casual) they collect data for, which could impact the relative importance of attributes. Wang et al. (2022) study the preferred product attributes for sustainable outdoor apparel and identify recycled fabric as the most important sustainable attribute.

Increasing Consumer Awareness. Additionally, there is a lack of research measuring the effect of increased awareness (or consumer education) on preferences and WTP for environmental attributes in fashion, and on its impact on retailers' profitability. Past research has focused on labels as a vehicle of information (e.g., [Turunen and Halme 2021](#)), versus directly engaging with consumers as a means to relay information (or educate). [McKie et al. \(2023\)](#) look at the drivers for participating in a take-back program for jeans, and find that providing generic information to consumers leads to the highest uptake in such a program.

Sustainable Operations and Fashion. The marketing literature concerned with measuring preferences and WTP has not yet connected to operational decisions. Conversely, research in OM focusing on sustainability in the fashion industry has taken a mostly analytical approach, studying various business models to reduce overproduction. [Long and Nasiry \(2022\)](#) and [Long and Gui \(2024\)](#) study the effect of the quick response model, and [Alptekinoğlu and Örsdemir \(2022\)](#) that of mass customization on production, durability, and leftover inventory, and how these could mitigate their related environmental impacts. [Alptekinoğlu et al. \(2023\)](#) experimentally assess how different points of customer involvement in mass-customization impacts time-to-disposal and WTP.

Our work brings together a robust integrated Marketing-OM perspective by combining an incentive-aligned choice-based conjoint (IA-CBC) experiment to measure preferences and WTP for attributes of fast fashion garments and prescriptive analytics (counterfactual simulation and optimisation balancing profit and environmental improvements) to study how the transition to less environmentally impactful production can be driven optimally, providing rich behavioural and managerial empirical insights. Our experimental approach, as opposed to prior studies, allows estimating real preferences and WTP, and has the advantage of being both multi-dimensional (multiple environmental attributes) and multi-level (varying levels) within each environmental attribute, addressing the current lack, both in the academic literature and industry, of measuring consumer preferences for different levels of environmental attributes (i.e., not only measuring preferences when an attribute is included or not). Furthermore, our experimental methodology includes a treatment (showing a short educational video) that measures the effect of increasing consumer awareness on consumer's preferences and WTP for environmental attributes as well as a retailer's profitability. To the best of our knowledge, our study is the first to rigorously measure preferences for multi-level environmental attributes in fast fashion, how they impact business profitability and environmental improvements, and how they relate to proposed regulations.

3. Experimental Methodology

This section presents the experimental methodology of our integrated marketing-operations framework (Steps 1 through 3 depicted in Figure A1 in the Appendix). To promote transparency in research and facilitate the reproducibility of our study, the project's Open Science Framework (OSF) page² contains supplemental information about the experiment (instructions, questions, etc.) as well as the programming code (in Python and R) used to generate the experimental design.

3.1. Design of the Conjoint Experiment

The goal of our experiment is to collect choice data to measure consumer preferences and willingness-to-pay for environmental attributes (RQ1). In our setting, an experiment is necessary since the environmental attributes and their levels are not yet all available on the market. Specifically, we leverage a choice-based conjoint (CBC) design with three alternatives plus a “no-choice” alternative where respondents make repeated trade-offs among varying sets of products defined by six attributes (three environmental and three base attributes) designed to be hard, leading to maximally informative choices. We design hyper-individualized estimation choice tasks following the method proposed by Liu and Tang (2015), each estimation set composed of 15 choice tasks, in addition to five validation tasks, for a total of 20 choice tasks per respondents. The same five fixed validation sets shown to all respondents, are dispersed over the course of the experiment (one validation set every 4 estimation set), and varied between respondents according to a Latin square design. Appendix EC.1.1 details the design of the estimation and validation sets. During the experiment we also collect answers to questions asked before and after the choice tasks. These additional data model the heterogeneity in consumer preferences.

Since the observed choices in the conjoint experiment are from hypothetical situations imposing no real consumption consequences for the respondents, resulting choice data may not fully convey their real preferences. To avoid biased preference estimates, our experimental design includes a truth-telling mechanism referred to as an incentive-aligned mechanism that enables us to unveil the true preferences, and most importantly, the real WTP (Schmidt and Bijmolt 2020). We use a Becker-DeGroot-Marschack (BDM)-based incentive-alignment mechanism proposed by Ding (2007). This mechanism has been extensively used in the literature (e.g., Park et al. 2008, Miller et al. 2011, Wlömert and Eggers 2016). Respondents received clear instructions at the beginning of the process and were shown a short video explaining the auction procedure (i.e., the incentive-alignment mechanism). They were then required to review a table defining all attributes and levels before proceeding with the experiment.

Stimulus. We conduct the conjoint experiment for a hypothetical sweatshirt category of a fast fashion brand describing the sweatshirts using six attributes each comprising five levels. Appendix EC.1 presents

²Follow [this link](#) to access the project's OSF page.

Table 2 Attributes and Levels

(a) Base Attributes		(b) Environmental Attributes	
Attribute	Levels	Attribute	Levels
<i>Bill of Materials</i>	100% cotton	<i>Recycled Content</i>	0% (virgin)
	75% cotton, 25% polyester		15% recycled content
	50% cotton, 50% polyester		30% recycled content
	25% cotton, 75% polyester		45% recycled content
	100% polyester		60% recycled content
<i>Type</i>	stylish	<i>Durability</i> (increase in lifespan)	None
	lounge		10% increase
	casual		20% increase
	active		30% increase
	essential		40% increase
<i>Price (£)</i>	10	<i>Circularity Design</i>	None
	25		designed for repairs
	40		designed for redistribution
	55		designed for upcycling
	70		designed for recycling

the example sweatshirt shown to respondents at the beginning of the choice experiment. Table 2 summarizes the attributes and their levels.

We focus on three *environmental* attributes³. *Recycled Content* represents how much fibre-to-fibre recycled content is incorporated in the sweatshirt. *Durability* represents the increase in lifespan (or wears) of a sweatshirt from using better fabric and more robust stitching. While durability could also be understood as higher quality, we argue both are highly correlated, with improvements in one directly leading to improvements in the other, thus motivating consumers to purchase because the product will last longer.⁴ *Circularity Design* refers to alternate pathways a garment can take to make the current linear model circular. We define Circularity Design as whether the product was designed with any end-of-life circularity pathway in mind, and the levels are defined based on four alternate pathways a garment can take instead of going to the land-fill or incinerator.⁵ These three environmental attributes were chosen specifically because of their relevance to the adopted EU Strategy.

To retain a balanced design and allow comparison with conventional product attributes, we also describe the sweatshirts on three *basic* attributes: *Bill of Materials (BoM)*, *Type*, and *Price*. We define levels of BoM to be representative of the offerings at most fast fashion retailers at the time of data collection: natural-only, synthetic-only and synthetic-natural blends typical of sweatshirt categories. We also include Type to

³Recall we refer to attributes that have the potential to reduce environmental impacts as environmental attributes.

⁴According to a [Cotton Inc. report on durability](#), a high percentage of consumers (84%) find it “important that the clothes they purchase are durable enough to last a long time,” “70% are likely to look for durability enhancement features to extend the life of the apparel they buy”, and “52% are willing to pay more for better quality apparel”, showing that consumers implicitly link higher quality with increased lifespan. This consumer expectation highlights the high perceived correlation between quality and durability.

⁵See [this link](#) for a helpful visualization of circularity in fashion.

capture various styles of sweatshirts, assuming levels are horizontally differentiated. Featuring this attribute enhances external validity. We set Price levels by noting current offerings of sweatshirt products from leading European fast fashion retailers. Appendix EC.1.2 presents brands studied and price ranges observed.

Increased Awareness Treatment. §1 alluded to the misalignment among parties in the retail ecosystem (consumers, retailers, and policymakers) and the profitability-environmental impact trade-off experienced by retailers blocking a switch to less impactful production. We study how increased consumer awareness, implemented by showing a short educational video to respondents, could help align the three parties towards a lesser environmental footprint (RQ3). In our experiment, nearly half of the respondents are shown a short video (about two and a half minutes) briefly presenting each environmental attribute and providing a high-level overview of how increased levels of each attribute can lower the environmental impacts of a fashion product.

Final Sample. We recruit 500 respondents through Prolific aged 18-45, residents of the United Kingdom, having recently shopped at a fast fashion retailer. We limit data collection to respondents aged 18 to 45 as they represent the majority of the consumer population for fast fashion (the chosen age range captures Gen Z (18-23) and Millennials (24-42)). Furthermore, we restrict our data collection to respondents living in the United Kingdom because of their proximity to Europe, and language.⁶ The experiment is carried out in Limesurvey, and the choice task data collection is done using Sawtooth software. Respondents are directed from Limesurvey to Sawtooth, and back to Limesurvey, automatically. Respondents are paid £6/hour for participating in the experiment that took on average 18.6 minutes per respondent to complete. Our final sample contains 489 responses (rejecting 11 proven incomplete): 231 respondents received the increased awareness treatment, while 258 did not.

3.2. Demand Model Estimation

We assume consumer choice to follow a Hierarchical Bayes Multinomial Logit model (HB-MNL), where we define consumer utility as the sum of all basic and environmental partworth utilities. Mathematically, the utility respondent $i = 1 \dots I$ attaches to alternative $j = 1 \dots J$ in choice set $k = 1 \dots K$ is written as:

$$u_{ijk} = \underbrace{\sum_{a=1}^{A^b} \sum_{l=1}^{L_a^b-1} \beta_{ial}^b x_{ijkal}^b}_{=u_{ijk}^{\text{base}}} + \underbrace{\sum_{a=1}^{A^e} \sum_{l=1}^{L_a^e-1} \beta_{ial}^e x_{ijkal}^e}_{=u_{ijk}^{\text{environ}}} + \epsilon_{ijk} \quad (1)$$

where superscripts b and e refer to basic and environmental attributes respectively, A^f corresponds to the number of attributes for $f \in \{b, e\}$, and L_a^f is the number of levels for attribute a of type f . β_{ial}^f corresponds to the partworth coefficient of respondent i for level l of attribute a and x_{ijkal}^f is an indicator variable equal to

⁶Because the United Kingdom speaks English, no translation of the experiment was necessary.

1 if alternative j presented to respondent i in choice set k has level l of attribute a , 0 otherwise. We assume level $l = 0$ as the base level for each attribute. In addition, we define the utility of the “no-choice” alternative in choice set k for respondent i as $u_{i0k} = \beta_{i0} + \epsilon_{i0k}$. In doing so, we follow [Haaaijer et al. \(2001\)](#), who show that failing to define the utility of the no-choice alternative (i.e., setting it to zero) can lead to biased partworth coefficient estimates. The random errors ϵ_{ijk} are assumed to follow a *Gumbell*(0,1) distribution. We assume the $(n \times 1)$ vector of individual-level partworth utilities, β_i , to be distributed according to a multivariate normal distribution:

$$\beta_i = \left[\beta_{i0}, \beta_{i11}^b, \dots, \beta_{iAb(L_a^b-1)}^b, \beta_{i11}^e, \dots, \beta_{iAe(L_a^e-1)}^e \right]^T \sim MVN(\Theta_1 z_i + \theta_2 ed_i, \Sigma) \quad (2)$$

where Σ represents a diagonal covariance matrix, and the population mean is a function of an $(m \times 1)$ vector z_i , containing an intercept and $m - 1$ individual characteristics we collect during the experiment (socio-demographics, self-reported measures of Recency, Frequency & Monetary value (SR-RFM), self-reported environmental consumption attitudes (SR-ECA)), and indicator variable $ed_i = 1$ if respondent i is in the experimental treatment group (i.e., shown the short educational video), $ed_i = 0$ otherwise. Finally, the outcome of each choice task is captured by y_{ijk} equal to 1 when respondent i chooses alternative j in choice set k , 0 otherwise. Appendix [EC.2](#) discusses data preprocessing and manipulations performed to the experimental data collected, plus a discussion on the Confirmatory Factor Analysis (CFA) performed on the SR-ECA data.

Estimation Approach. We estimate the parameters for the above HB-MNL model with Hamiltonian Monte Carlo using Rstan ([Stan Development Team 2019](#)). We run the model for 4,000 warm-up iterations, and 4,000 sampling iterations over four chains, resulting in 16,000 draws for posterior parameter inference. Diagnostics confirm the chains converged and mixed well. We use the estimation choice sets (15 per respondent) for model estimation, and assess the predictive validity of the estimated model on the validation choice sets (same five for each respondent). Table [EC.3](#) summarises the model validity, showing that our model has both high in-sample and out-of sample fit, and conclude the model fit is adequate.

3.3. Estimating Willingness-to-Pay

While partworth utilities reveal consumer preferences for different environmental attributes, they do not immediately provide researchers and managers with the monetary value associated with these preferences. Therefore, we also estimate the willingness-to-pay (WTP) for different levels of the environmental attributes by adapting the methodology proposed by [Kohli and Mahajan \(1991\)](#) and extensively used in the literature (e.g., [Miller et al. 2011](#), [Schmidt and Bijmolt 2020](#)), enabling a retailer to assess whether the value created by investing in these attributes (indicated by WTP) justifies the cost. The method compares two products (baseline versus focal product), the only unknown being the price of the focal product, and finds the minimal

price for the focal product that makes the consumer indifferent between the two products. This relationship can be written as:

$$u_{ij|-p} + \nu_{ij}(p) \geq u_i^* \quad (3)$$

where $u_{ij|-p}$ corresponds to the total utility, excluding the utility of price, respondent i attaches to product j , $\nu_{ij}(p)$ corresponds to the utility of price p respondent i attaches to product j , and u_i^* denotes the total utility of a baseline for respondent i against which we compare product j . A pseudo-code of the WTP calculation is presented in Appendix EC.5.

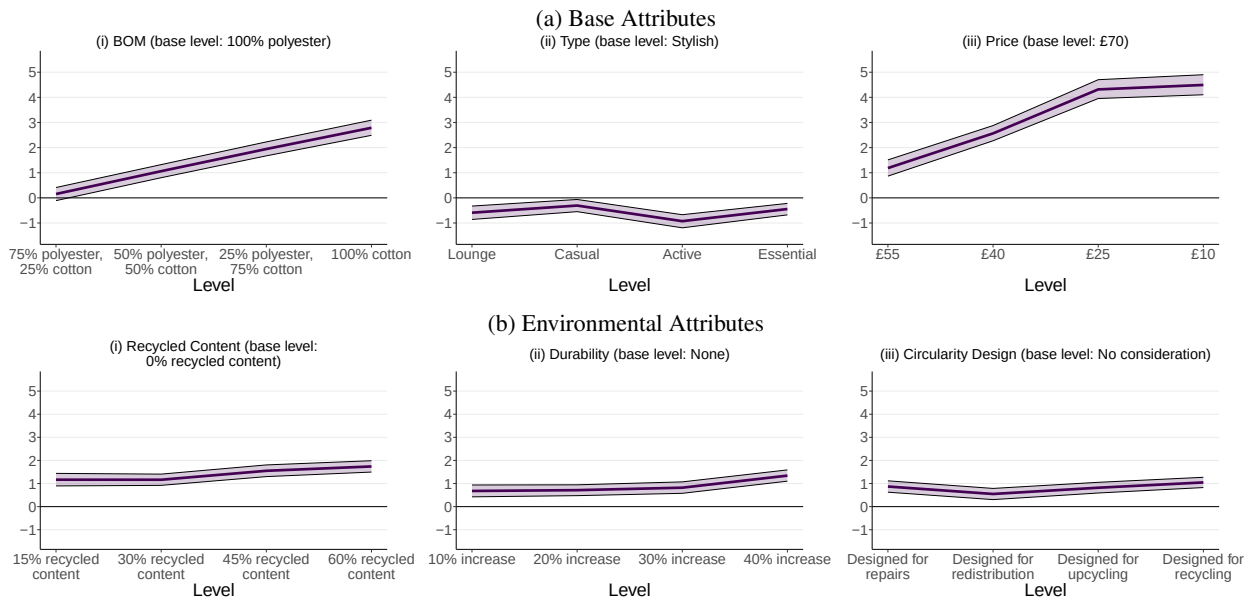
4. Preferences and Willingness-to-Pay Results

This section presents the main results from our consumer preference measurement and willingness-to-pay study. We begin by reporting partworth and willingness-to-pay (WTP) estimates for the control group—that is, respondents who were not shown the short educational video. We then assess the impact of the video by comparing preferences and WTP between the treatment and control groups.

4.1. Consumer Preferences and WTP—the Status Quo

Partworth Estimates. Figure 2 summarises the mean (across posterior draws) average (across respondents) posterior distribution of partworth utilities for base and environmental attributes respectively. We make the following key observations.

Figure 2 Posterior Partworth Utility Distributions



Note. Bold lines indicate the posterior mean (across posterior draws) average (across respondents) partworth utilities. Shaded areas indicate the 95% credible intervals. For all attributes, the utility of the first (base) level is set to 0 for identification purposes, all other partworth utilities being relative to it. The mean average no-choice utility is 6.26.

Environmental attributes matter, but they are not the primary drivers of choice. Figure 2, panel (b), illustrates the partworth utilities for the environmental attributes. Note that the utility of the baseline level—defined as “no sustainability consideration”—is set to zero. Importantly, the partworth utilities for all levels beyond this baseline are consistently above zero, indicating that consumers positively value sustainability features. Moreover, the range of utilities across all levels of each attribute—the difference between the highest and lowest partworth values, including the baseline—serves as a proxy for the relative importance of the attribute, a point we return to later. The utility ranges for the three environmental attributes are similar in magnitude, though the range for Recycled Content (panel b-i) is slightly higher than those for Durability (panel b-ii) and Design for Circularity (panel b-iii). Notably, the smallest individual utility is observed for the “Design for Redistribution” level. This may reflect consumer skepticism, potentially rooted in concerns that such schemes are primarily resale opportunities allowing retailers to extract double profits from consumers (McKie et al. 2023).

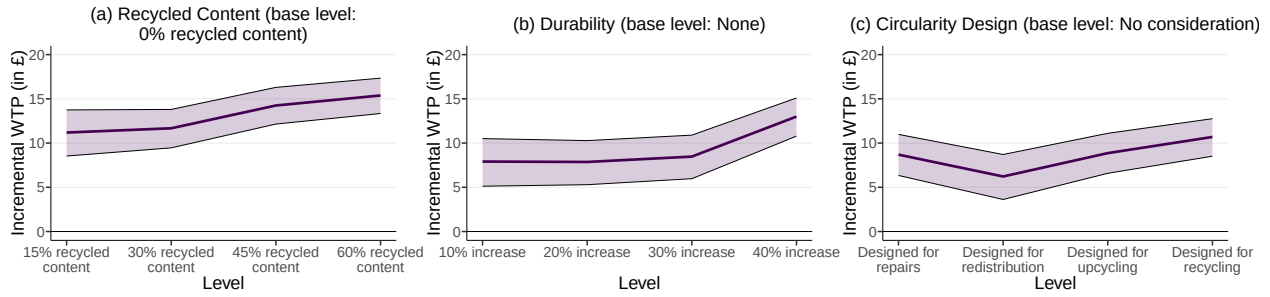
Consistent with the academic literature, we find Price to be the primary driver of choice. The steep slope of the partworth utilities across price levels in Figure 2, panel (a-iii), indicates substantial price sensitivity. However, the curve plateaus at the lower end of the price range, suggesting that consumers are largely indifferent between the two lowest price points. We also find BoM to be a significant driver of choice. As one of the few readily available and widely understood indicators of garment quality, BoM likely derives its high utility range from its perceived informational value to consumers. Figure 2, panel (a-i), shows a clear increase in utility as the composition shifts from pure synthetic fibres to pure natural fibres.

Utility for Environmental Attributes is a Step Function. Figure 2 panel (b) shows that while respondents in the control group are generally insensitive to higher levels of the environmental attributes, any level of environmental attributes is preferred to none. This reveals a discontinuous jump in utility at the zero-attribute level—a step-function pattern with important economic implications discussed in §6.

Inconsistencies Regarding Durability. A key finding of our experiment is the discrepancy between the revealed importance of Durability and what respondents report in follow-up questions. As shown in Figure 2, panel (b), the partworth utilities suggest that Recycled Content carries slightly greater weight than Durability in driving choice. However, when asked to directly rank the three environmental attributes by relevance, 63% of respondents identify Durability as the most important (followed by Recycled Content and Circularity Design; see Appendix EC.1.3, Table EC.2). This contradiction between indirect (revealed) and direct (stated) preferences underscores a tension between consumer beliefs and actual decision-making—a point we return to in §6.

Willingness-to-Pay. Figure 3 shows the posterior distribution of the incremental WTP (in £), for each environmental attribute. The shapes of these distributions closely mirror those of the mean partworth utilities in Figure 2, panel (b), but are easier to interpret as they represent monetary values.

Figure 3 Posterior Distributions of the Incremental Willingness-to-Pay



Note. The incremental WTP corresponds to the difference in monetary value respondents are willing to pay between a product with no environmental attributes (i.e., the base level for all three focal environmental attributes), and a product with a given level of a given environmental attribute (e.g., 30% recycled content). Bold lines indicate the mean (across posterior draws) average (across respondent) posterior partworth utilities. Shaded areas indicate the 95% credible interval.

Figure 3 reveals that, despite appearing relatively insensitive to environmental attributes in the choice tasks, consumers assign them meaningful monetary value. This suggests that when two alternatives are comparable on core attributes, consumers may favor the more sustainable option. Notably, the WTP distributions for Recycled Content and Circularity Design exhibit a step-to-linearly-increasing pattern. This indicates that offering more than minimal levels of these attributes may yield higher returns on investment (ROI)—a contrast to the more modest differences implied by the partworth estimates. We explore this tension between consumer WTP, implementation costs, and current monetary incentive policies in §6. Furthermore, [Rausch et al. \(2021\)](#) estimate willingness to pay (WTP) as discount equivalents of 30% for Durability and 19% for Recycled Content, which correspond to £10.35 and £6.56, respectively, for a £34.50 sweatshirt. In comparison, our WTP estimates fall within a similar range but indicate a higher valuation for Recycled Content than for Durability. This alignment, along with the observed preference reversal, lends additional face validity to our results.

4.2. Impact of the Awareness-Increasing Treatment on Preferences and WTP

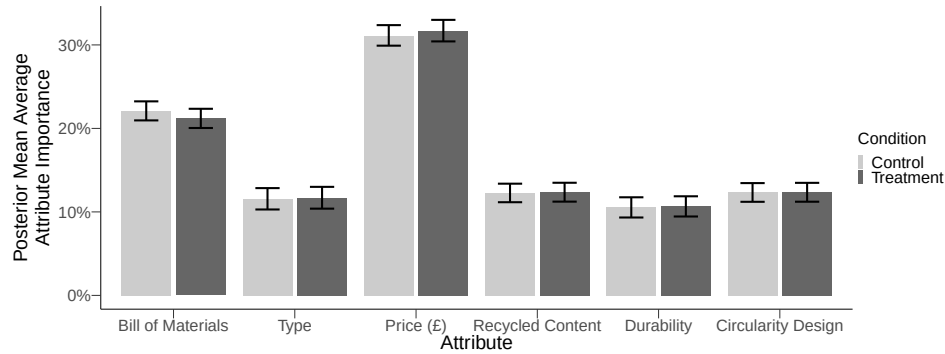
We now shift our focus to answering our third research question by characterizing the impact of increased consumer awareness, stimulated by showing our short educational video, on respondent preferences and willingness-to-pay for environmental attributes.

Attribute Importance. We compute the attribute importance as the ratio of the partworth utility range for a focal attribute over the sum of partworth ranges of all attributes ([Cattin and Wittink 1982](#), [Eggers et al. 2021](#)). Formally, for respondent i and posterior draw s , we compute the attribute importance of attribute a of type $f \in \{b, e\}$, $IMP_{ia,s}^f$, as:

$$IMP_{ia,s}^f = \frac{\max_{l \in \{0, \dots, L_a^f - 1\}} \beta_{ial,s}^f - \min_{l \in \{0, \dots, L_a^f - 1\}} \beta_{ial,s}^f}{\sum_{f' \in \{b, e\}} \sum_{a'=1}^{A^{f'}} \left(\max_{l \in \{0, \dots, L_{a'}^{f'} - 1\}} \beta_{ia'l,s}^{f'} - \min_{l \in \{0, \dots, L_{a'}^{f'} - 1\}} \beta_{ia'l,s}^{f'} \right)} \quad (4)$$

where $\beta_{ial,s}^f$ is the posterior draw s for the partworth utility that respondent i attaches to level l of attribute a of type f . We focus on the average attribute importance across respondents and posterior draws. Figure 4 contrasts the posterior distributions of the average (across respondents) attribute importance for the control group with those for treatment group.

Figure 4 Posterior Mean Average Attribute Importance (Control vs. Treatment)



Note. Mean refers to the posterior mean, and average to the average across respondents in the corresponding condition. Error bars represent the 95% posterior credible interval.

Our results reveal small differences in attribute importance between the control and treatment groups. Figure 4 reveals a decrease in attribute importance for Bill of Materials, accompanied by modest increases in attribute importance for all environmental attributes and Price. The largest increase in environmental attributes happens for Durability, while increases for Recycled Content and Circularity Design are similar. The increase in Price can be interpreted in two different ways. On the one hand, this increase in importance can signal consumers are less willing to pay for environmental attributes. On the other hand, this increase in importance can make the no-choice option more attractive, lowering consumption. We show later that this is indeed a behaviour we observe. We quantify those differences by measuring the Sample Average Treatment Effect (SATE, see Appendix EC.4). Our results confirm a significant decrease in the importance of BoM (4.3 percentage points with 96.4% of the distribution density below 0), but marginal increases in the importance of the environmental attributes.

Willingness-to-Pay. We also see modest increases in willingness-to-pay for environmental attributes with the treatment (see Appendix EC.5.1 Figure EC.4).

No-Choice Partworth Utility. Importantly, we observe an increase in the no-choice partworth utility between the control and treatment group. Because the no-choice utility is only meaningful relative to the other attributes' utilities, we calculate the relative no-choice utility as the ratio of the no-choice utility to the total range of partworth utilities (the denominator of eq. (4)), providing us with a normalised measure of the importance of the no-choice alternative for each group.

We quantify the difference between groups by measuring the SATE and find a significant increase of the relative no-choice utility with the treatment from 0.30 in the control group to 0.31 in the treatment group, with 93.8% of the distribution density above 0. This key observation has the following implication: increased awareness makes consumers more critical of the offering, and more likely to not choose (i.e., purchase) any product. This latter key result can be interpreted differently by advocacy groups and retailers. We discuss this in more detail in §6.

Concluding Remark. While at first glance the treatment effects may appear modest—raising questions about the value of increased consumer awareness given the costs of engagement—our findings suggest otherwise. Even small shifts in preferences and willingness-to-pay, alongside a heightened likelihood of opting out, indicate that increased awareness makes consumers more discerning and environmentally attentive. As we show in Section §5, these subtle behavioural changes meaningfully improve the trade-off between profitability and environmental impact, underscoring the strategic value of targeted awareness interventions.

5. The Profitability-Environmental Impact Trade-Off

We now shift our focus to the economic implications of our behavioural findings by analysing a joint product line selection and pricing problem, subject to environmental improvement restrictions on the three environmental attributes.

5.1. Joint Product Line Selection & Pricing (jPLSP) Problem with Environmental Improvement Constraints

We focus on vertically integrated retailers making their own production decisions (e.g., Zara, H&M). Such retailers face two interdependent challenges: 1) determining what products to include in their assortment, and 2) setting prices for these products. We formalize these challenges into a joint Product Line Selection and Pricing (jPLSP) problem maximizing profitability, subject to constraints on the assortment's environmental impact improvements. This optimization problem integrates information from the three parties in the fashion system: consumer product valuations (which may vary with increased awareness), retailer costs and pricing structure, and environmental improvement restrictions imposed by policymakers (or self-motivated retailers).

Formulating jPLSP. Let $N = \{1, \dots, |N|\}$ represent the universe of product versions a retailer can include in its assortment of five products, with each version corresponding to a specific combination of levels of Bill of Materials (BoM), Recycled Content (RC), Durability (D), and Circularity Design (CD), $|N| = 5^4 = 625$. We assume consumers are indifferent to Type, and thus use a single average value for the Type utility. Additionally, we assume the retailer must select the price level from a set $R = \{1, \dots, |R|\}$

of options, for each included product. The joint Product Line Selection & Pricing (jPLSP) optimization problem can then be expressed as follows:

$$(jPLSP) \quad \max_x \sum_{q \in N} \sum_{r \in R} (p_{qr} - c_q) \mathbb{E}_\beta [D_{qr}(x)] \quad (5a)$$

$$s.t. \quad \sum_{r \in R} x_{qr} \leq 1 \quad \forall q \in N \quad (5b)$$

$$\sum_{q \in N} \sum_{r \in R} x_{qr} = 5 \quad (5c)$$

$$\sum_{q \in N} \sum_{r \in R} IMP_q^{RC} x_{qr} \geq t^{RC} \quad (5d)$$

$$\sum_{q \in N} \sum_{r \in R} IMP_q^D x_{qr} \geq t^D \quad (5e)$$

$$\sum_{q \in N} \sum_{r \in R} IMP_q^{CD} x_{qr} \geq t^{CD} \quad (5f)$$

$$x_{qr} \in \{0, 1\} \quad \forall q \in N, \forall r \in R \quad (5g)$$

Decision variables. Binary decision variables x_{qr} are equal to 1 if version q at price level r is included in the product line, 0 otherwise.

Objective function. The objective function is the sum-product across all included products of (a) the gross profit margin (p_{qr} , the price of version q when level r is chosen, minus c_q , the cost of version q) and (b) the expected demand for the corresponding version q at price level r given the full assortment x , $\mathbb{E}_\beta [D_{qr}(x)]$, where $D_{qr}(x) = \frac{v_{qr} x_{qr}}{v_0 + \sum_{q'} \sum_{r'} v_{q'r'} x_{q'r'}}$ is the random choice share for product version q at price r given assortment x and the expectation is taken with respect to $\beta = \{\beta_i\}_{i=1, \dots, I}$.

Constraints. Constraints (5b) ensure there can be only one version at one price selected, and Constraint (5c) limits the size of the assortment to five product versions. We choose five to align closely with our experimental setting (a three-alternative CBC design plus the no-choice option), while providing enough products to better study the distribution of optimal attribute levels. Constraints (5d) to (5f), referred to as environmental improvement constraints, force the sum of the impact improvement scores of each environmental attribute to meet or exceed a certain threshold t^e , $e \in \{RC, D, CD\}$ where IMP_q^e , $e \in \{RC, D, CD\}$ represents the impact improvement scores for each version q for each environmental attribute e . For all three environmental attributes, level 0 (i.e., the baseline) is assigned an impact improvement score of 0, as it represents the status quo (i.e., no improvements). Each subsequent level is given an increasing impact improvement score (e.g., level 1 is assigned a score of 1, level 2 is assigned a score of 2, etc.). For Circularity Design, each level is uniformly assigned a score of 1 (i.e., all circularity pathways are rewarded equally).

Price levels. We construct product version price sets by first calculating the mean average WTP for each version, rounding it to the nearest £5, and then generating five price levels: the rounded WTP, two levels

below (WTP-£10, WTP-£5) and two levels above (WTP+£5, WTP+£10). Product costs are defined by applying a cost multiplier to each attribute level, which is then multiplied by the cost of a base product (i.e., a version with no environmental attributes, estimated at £13.80).⁷ Table 4 provides an overview of the cost multipliers, while Appendix EC.7 outlines the details of how each multiplier was operationalized.

Table 4 Cost Multipliers

	Level 1 (Status quo)	Level 2	Level 3	Level 4	Level 5
Bill of materials	0.80	0.90	1.00	1.10	1.20
Recycled content	1.00	1.20	1.35	1.60	2.00
Durability	1.00	1.15	1.25	1.35	1.45
Circular design	1.00	1.10	1.13	1.05	1.05

Solving jPLSP Efficiently. Solving jPLSP is challenging due to the nonlinear, non-concave, and stochastic nature of the objective function. To address these complexities and make the problem tractable, we use two methods: (1) demand approximation and (2) mixed integer linear programming (MILP) reformulation.

Demand approximation. We approximate the expected demand function by combining (a) an approximate Bayes method and (b) posterior post-hoc segmentation (Green and Krieger 1991). For the former we use the *plug-in* approach (Morris 1983), in which the integration over the posterior distribution of model parameters is replaced by a single evaluation at the posterior mean of the partworth estimates.⁸ For the latter, we apply a two-step cluster analysis on the individual-level posterior mean partworth vectors, grouping individuals into M distinct segments. In the first step, we use hierarchical clustering with the Ward linkage (Ward 1963) to determine the optimal number of clusters. We then use K-Means clustering to optimise the segmentation of each respondent. We find four segments (see Appendix EC.8 for the dendrogram) and obtain from this procedure mean average partworth estimates per segment $\beta^m = [\beta_0^m, \beta_{al}^m, \dots, \beta_{al}^m, \dots]^T$ with which we then calculate the average choice probabilities from posterior means for each segment m , and weigh these probabilities per relative cluster size π_m . The result is a parametrization of the stochastic demand function as a mixture of Multinomial Logit models (MNLs).

Reformulation of the optimisation problem. Second, we reformulate the problem as a mixed integer linear programming (MILP) by introducing auxiliary variables y_{qrm} that correspond to the choice probability of segment m choosing version q at price level r from assortment x . The combination of the mixture of

⁷We assume an average price to be £34.50, and assume a 60% gross margin. As reported recently by CNBC, Zara's gross margin is 58%.

⁸This method is widely used for improving computational efficiency has been adopted by Sawtooth Software, an industry leader in market research and choice-based simulation tools, for their choice simulators.

MNLs and the MILP formulation yields a tractable optimization problem that can be efficiently solved using commercial solvers. We express the resulting reformulation as:

$$(jPLSP - MILP) \quad \max_{x,y} \sum_{m \in 1, \dots, M} \pi_m \left[\sum_{q \in N} \sum_{r \in R} (p_{qr} - c_q) y_{qrm} \right] \quad (6a)$$

$$s.t. \quad \sum_{r \in R} x_{qr} \leq 1 \quad \forall q \in N \quad (6b)$$

$$\sum_{r \in R} \frac{y_{qrm}}{v_{qrm}} \leq \frac{y_{0m}}{v_{0m}} \quad \forall q \in N, \forall m \in 1, \dots, M \quad (6c)$$

$$\sum_{q \in N} \sum_{r \in R} x_{qr} = 5 \quad (6d)$$

$$\sum_{q \in N} \sum_{r \in R} \frac{y_{qrm}}{v_{qrm}} = 5 \frac{y_{0m}}{v_{0m}} \quad \forall m \in 1, \dots, M \quad (6e)$$

$$\sum_{q \in N} \sum_{r \in R} IMP_q^{RC} x_{qr} \geq t^{RC} \quad (6f)$$

$$\sum_{q \in N} \sum_{r \in R} IMP_q^D x_{qr} \geq t^D \quad (6g)$$

$$\sum_{q \in N} \sum_{r \in R} IMP_q^{CD} x_{qr} \geq t^{CD} \quad (6h)$$

$$y_{0m} + \sum_{q \in N} \sum_{r \in R} y_{qrm} = 1 \quad \forall m \in 1, \dots, M \quad (6i)$$

$$x_{qr} \geq y_{qrm} \quad \forall q \in N, \forall r \in R, \forall m \in 1, \dots, M \quad (6j)$$

$$y_{qrm} v_{0m} \leq y_{0l} v_{qrm} \quad \forall q \in N, \forall r \in R, \forall m \in 1, \dots, M \quad (6k)$$

$$x_{qr} \in \{0, 1\} \quad \forall q \in N, \forall r \in R \quad (6l)$$

where $v_{qrm} = \exp(u_{qrm})$ and $v_{0m} = \exp(u_{0m})$. Since we are modelling a mixture of multiple consumer segments, we retain decision variable x to enforce the retailer to select the same assortment for *all* consumers. Forcing constraints (6j) ensure that a product version cannot be offered to segment m unless it is chosen globally. The full derivation of the MILP formulation can be found in Appendix EC.6.

Contrasting our approach to Latent Class Analysis. We estimate continuous individual-level preferences using Hierarchical Bayes (HB) and apply post-hoc segmentation, rather than using a Latent Class model during estimation. This approach offers several advantages. First, HB captures the full extent of heterogeneity by assuming a continuous distribution of preferences, whereas finite mixture models can only approximate this with a small number of discrete mass points, often inadequately so in practice (Allenby and Rossi 1998). Moreover, Latent Class models constrain individual-level posteriors to the convex hull of segment-level preferences, leading to underestimation of true heterogeneity (Allenby and Rossi 1998). In contrast, HB provides richer individual-level estimates that can be flexibly segmented post hoc. Importantly,

HB also allows for the straightforward incorporation of observable covariates, such as sociodemographics or experimental treatments, as drivers of heterogeneity in preferences (Allenby and Ginter 1995), which improves explanatory power and interpretability. In sum, the separation of estimation and segmentation allows for flexibility in individual-level preference estimation as well as aggregation driven by the specific decision context.

Counterfactual Simulation. In solving jPLSP-MILP, we leverage our experimental manipulation and the posterior draws from our Bayesian inference approach to conduct a counterfactual simulation that explicitly accounts for heterogeneity in individual-level preferences arising from both differences in individual-level characteristics (z_i) and exposure to the treatment (ed_i). This simulation is central to our analysis of economic implications, as it enables us to assess the trade-offs between profitability and environmental impact for the same set of individuals across two different scenarios.

Formally, for each respondent i in the treatment group, we calculate counterfactual preferences as if they had been in the control group for a given posterior draw s as $\beta_{i,s}^{ed^*=0} = \beta_{i,s} - \theta_{2,s}$, while their observed preferences under treatment remain $\beta_{i,s}^{ed^*=1} = \beta_{i,s}$. For each respondent i in the control group, we apply the reverse transformation: $\beta_{i,s}^{ed^*=0} = \beta_{i,s}$, with the corresponding counterfactual under the treatment being $\beta_{i,s}^{ed^*=1} = \beta_{i,s} + \theta_{2,s}$. This counterfactual simulation produces two alternative scenarios: one in which no respondent received the treatment ($ed^* = 0$, referred to as the *status quo* scenario), and one in which all respondents were exposed to the treatment ($ed^* = 1$, referred to as the *increased awareness* scenario).

We perform the segmentation on average partworth utilities of consumers in each counterfactual scenario and confirm the segmentation is the same between scenarios. This approach allows consistent clustering of respondents as a function of the treatment.

5.2. Consumer Typology

Before moving to the optimisation results, we further put our behavioural insights into perspective by presenting the results of our segmentation analysis. The characterization of each segment allows for a richer understanding of the differences between consumer segments in our dataset, and the inherent compromise they bring when retailers try to maximise the welfare of all segments and improve their environmental footprint. Table 5 presents the segment-specific attribute importance.

Segment overview. Following the results of the hierarchical segmentation, we identify two primary clusters: *traditional-leaning* and *environmentally-leaning*. Each of these further divides into two sub-segments. The traditional-leaning cluster consists of (*price-sensitive* and *traditional*) segments, characterized by low importance placed on environmental attributes and high emphasis on price. In contrast, the environmentally-leaning cluster comprised the (*price-agnostic* and *environmentally-conscious*) segments, for whom environmental attributes hold the highest importance, while price plays a relatively minor role. Notably, the

Table 5 Segment-Specific Attribute Importance and Prevalance of Selecting the No-Choice Option

	Traditional-Leaning		Environmentally-Leaning		Overall
	Price-Sensitive (N=157, 32.1%)	Traditional (N=123, 25.2%)	Price-Agnostic (N=113, 23.1%)	Env.-Conscious (N=96, 19.6%)	
Attribute Importance					
Bill of Materials	19.8%	28.0%	31.3%	18.5%	24.5%
Type	6.8%	12.6%	8.2%	4.5%	6.9%
Price	48.1%	40.6%	23.1%	34.0%	39.1%
All Environmental Attributes	25.3%	18.8%	37.3%	42.9%	29.5%
- Recycled Content	10.1%	5.8%	15.0%	18.6%	11.7%
- Durability	9.9%	8.0%	9.6%	12.8%	10.6%
- Circularity Design	5.3%	5.0%	12.7%	11.5%	7.2%
No-Choice Proportion	0.33	0.28	0.20	0.28	0.28

Note. Table EC.5 in Appendix EC.8 describes the segments in terms of sociodemographics, condition assignment, and choice behavior. The no-choice proportion refers to the proportion of choices made by the segment that involved the selection of the no-choice option.

environmentally-leaning segments account for a smaller portion of the sample than the traditional-leaning segments, which is consistent with expectations based on prior research and market trends.

Attribute Trade-offs. As shown in Table 5 the relative importance of price compared to environmental attributes varies widely across segments. For instance, among *price-sensitive* respondents, price is up to 10 times more important than Circularity Design. Conversely, for *price-agnostic* and *environmentally-conscious* respondents, the ratio of price to Recycled Content importance drops to approximately 1.5. These disparities highlight the substantial challenge retailers face in designing product assortments that satisfy the diverse preferences across consumer segments.

No-Choice Proportion and Segment Motivation. The *price-sensitive* and *environmentally-conscious* segments exhibit the highest proportion of no-choice selection (0.33 and 0.28 respectively) within their respective cluster, indicating a greater likelihood of foregoing a purchase when offerings fail to meet their expectations. However, their motivations differ: price-sensitive consumers may be deterred by the lack of affordability or discounts, while environmentally-conscious consumers are likely influenced by the absence of sustainability features.

Unique Features of the Price-Agnostic Segment. A distinctive trait of price-agnostic respondents is the high importance they assign to the Bill of Materials (BoM). This, combined with their strong valuation of environmental attributes, suggests that these consumers are particularly attentive to what a product is made of—rather than its price or style. Additionally, they score the highest across most of the SR-ECA items, further underscoring their engagement with sustainability-related values.

Shopping Behavior Patterns. Both *price-agnostic* and *environmentally-conscious* respondents score highest on the recency item (see Appendix EC.8, Table EC.5) and relatively high on frequency, suggesting that environmentally-leaning consumers are active and regular shoppers. In contrast, the traditional segment

scores lowest on both recency and frequency, indicating more limited engagement in fashion consumption. Interestingly, some *price-sensitive* respondents show elevated importance for certain environmental attributes. This suggests a potential openness to sustainability, particularly when it aligns with immediate, tangible benefits—such as promotions or cost savings.

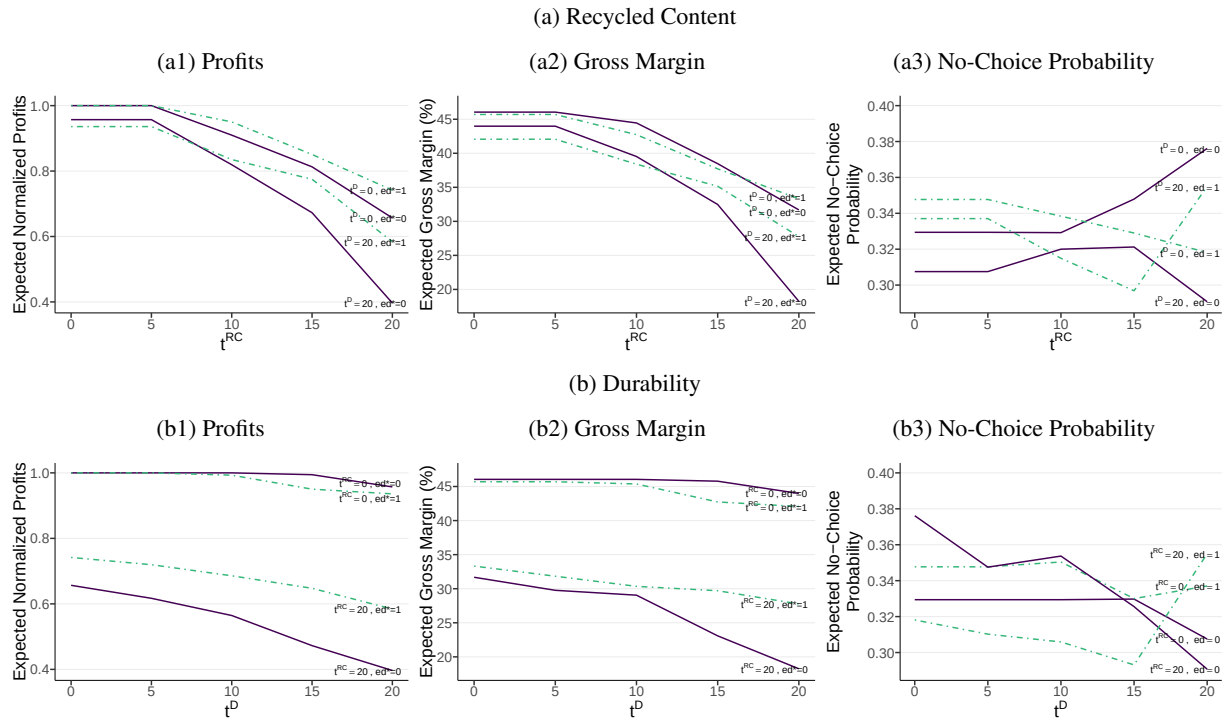
Sociodemographic Correlates. We observe that *environmentally-leaning* segments are composed of slightly more women than men, consistent with previous findings (e.g., [White et al. 2019](#)). In contrast, education level, which is often positively associated with sustainable consumption ([White et al. 2019](#), [Global Fashion Agenda and BCG 2019](#)), is in our sample more indicative of *price-sensitive* tendencies. Finally, self-reported socioeconomic status (SES) differs between clusters: environmentally-leaning respondents tend to position themselves lower on the SES scale compared to traditional-leaning respondents, adding nuance to the relationship between environmental concern and consumption capacity.

5.3. Impact of Environmental Improvement Restrictions and Treatment on Profitability

To assess the impact of environmental improvement constraints on profitability, we iteratively solve jPLSP-MILP by progressively increasing the thresholds for Recycled Content t^{RC} , Durability t^D and Circularity Design t^{CD} . Specifically, we consider five threshold levels for both for Recycled Content and Durability, $\{t^{RC}, t^D\} \in \{0, 5, 10, 15, 20\}^2$, and six threshold levels for Circularity Design, $t^{CD} \in \{0, 1, 2, 3, 4, 5\}$. This results in a total of 150 ($= 5^2 \cdot 6$) model instances that we solve for each counterfactual scenario. For each optimization instance, we evaluate the performance of the resulting optimal assortment by computing the expected profitability metrics (profits, gross margins, no-choice probability) using the ground truth utility data (i.e., for each respondent and each posterior draw under the corresponding counterfactual scenario).

The interaction between the thresholds for Recycled Content, Durability, and Circularity Design defines the profitability-environmental impact trade-off frontier. Figure 5 focuses on the interaction between Recycled Content and Durability, as our results indicate that Circularity Design is selected at a positive level in every optimal product configuration, making constraint (6h) non-binding in all but the highest threshold. Each panel in the figure presents two curves: one corresponding to the scenario with a minimal improvement threshold ($t^e = 0$) on the non-focal attribute, and one corresponding to the maximal improvement threshold ($t^e = 20$) on the non-focal attribute. We refer to these as the upper and lower limits respectively.

The Trade-off between Profitability and Environmental Performance. Focusing on the results for the status quo scenario ($ed^* = 0$; represented by the full purple lines), we find that constraints on Recycled Content have the strongest impact on the trade-off frontiers. This is evident both in the sharp decline in expected profits and gross margins as improvement thresholds increase (panels (a1)-(a2)), and in the wide gap between the upper and lower limits (panels (b1)-(b2)). Largely driven by the non-linear cost structure, requiring retailers to offer higher levels of Recycled Content proves particularly punitive, compelling them

Figure 5 Trade-Off Frontiers ($ed^* = 0$ vs. $ed^* = 1$)

Note. (1) Profits: Profits are normalised by the expected profit at $t^{RC} = t^D = t^{CD} = 0$ and $ed^* = 0$. (2) Gross Margin: Gross margin for each optimal product version is computed as $(price - cost)/price$. (3) No-Choice Probability. In all panels, the bold line indicates results for the status quo scenario ($ed^* = 0$), while the dot-dashed line indicates results for the increased awareness scenario ($ed^* = 1$).

to make tougher trade-offs across other product attributes. In contrast, constraints on Durability exhibit only effects on the trade-off frontiers. The nearly flat relationship between Durability and expected profits and gross margins (panels (b1)–(b2)) reflects its high utility-to-cost ratio—making it optimal to include higher levels of Durability even when improvement thresholds are low. For instance, even under the baseline constraint $t^D = 0$, the optimal product versions feature an average improvement score of at least $IMP^D \geq 2$. The effects on no-choice probability further differentiate the impact of these constraints. For Recycled Content, we observe a diverging pattern: as thresholds increase, the rising costs of compliance force retailers to compromise on Durability or the Bill of Materials (BoM), resulting in lower overall product utility and a corresponding increase in no-choice probability. Conversely, for Durability, the relationship is more convergent. The upper limit remains flat or decreases slightly, as increasing Durability thresholds does not require significant trade-offs with other attributes—yielding product versions that maintain or improve consumer utility and thus reduce no-choice probability.

We identify three key observations regarding the trade-off between profitability and environmental improvements.

Retailers Can Already Improve without Impacting Profitability. If environmental improvement constraints are imposed sequentially (i.e., one at a time) rather than simultaneously, our analysis shows that retailers

can already improve their assortment by incorporating less impactful products with no loss in bottom line. Under the status quo scenario, Figure 5 reveals no differences in expected profits, gross margin, or no-choice probability for thresholds up to $t^{RC} = 5$ and $t^D = 10$, when $t^D = 0$ and $t^{RC} = 0$ respectively.

Circular Design Should Always Be Offered. Because of their high utility-to-cost ratio, we find that it is almost always optimal for retailers to offer some level of Circularity Design, even without any constraints on t^{CD} . Exceptions arise only when constraints on either t^{RC} or t^D , or both, are high.

Trade-Offs Are Shaped by the Cost Structure. Our results also suggest that the impact of environmental improvement constraints on profitability is a function of the cost structure of the associated attribute. Since the utility for Recycled Content and Durability as a function of their levels is similar (see Figure EC.3 panels (b-i) and (b-ii)), Figure 5 suggests that an exponential cost structure (e.g., Recycled Content) results in a trade-off frontier that quickly decreases and follows the shape of the cost structure. On the other hand, for a linear cost structure (e.g., Durability), we see the trade-off frontier slowly decreasing. A direct implication of this observation is that had we assumed a linear cost structure for Recycled Content, the decrease in the trade-off frontier would have been more modest than shown, implying that our results might be conservative in cases where the cost structure is linear.

Impact of Increasing Awareness on the Profitability Trade-Off, Consumer Choice, and Retailer Strategy. We now focus on the results from the increased awareness scenario ($ed^* = 1$; represented by the dashed green lines) and its differences from the status quo scenario. Figure 5 shows the treatment helps to flatten the trade-off frontier for Recycled Content. Whereas the expected profits start at the same level for the upper limit, and even lower for the lower limit, compared to the status quo scenario, we see the expected profits in the increase awareness scenario to be substantially higher—13.0 and 47.3 percentage points respectively—at $t^{RC} = 20$. Our results also show the lower limit of the profit frontier (panel (a1)), and both the upper and lower limit of the gross margin frontier (panel (a2)) in the increased awareness scenario to start below that of the status quo scenario, but crossing over quickly. This is because while the short video increases overall WTP, it also pushes the retailer to include, on average, higher levels of environmental attributes from the outset, which comes at an additional cost that cannot be offset by the relatively small increase in WTP. We note that increased awareness has the same impact on gross margin results. Similarly, we see for Durability the upper limit of the expected profits (panel (b1)) and expected gross margins (panel (b2)) to be lower in the increased awareness scenario at $t^{RC} = 0, t^D = 20$ with drops of 2.2 and 4.7 percentage points respectively. These results further support our earlier behavioural insight that Recycled Content is the more salient environmental attribute.

We share three additional key observations on the impact of increased awareness on retailer profitability and decisions in three key areas: the threshold for profitability, consumer selectivity (as reflected in the no-choice probabilities), and the composition of the Bill of Material.

Increased Awareness Shifts the Probability Threshold. We first observe a shift in the threshold for profitability, defined as the inflection point in panels (a1) and (b1) of Figure 5. Under the status quo scenario, profitability begins to decline at $t^{RC} = 5$ and $t^D = 14$. In contrast, under the increased awareness scenario, these inflection points move to $t^{RC} = 8$ and $t^D = 9$. Because the treatment increases consumer preference for both Recycled Content and Durability, this shift suggests the treatment then enables the retailer to offer higher baseline levels of Recycled Content. However, this comes with a trade-off: a reduction in Durability levels to manage the increased cost of providing more recycled content.

Increased Awareness Makes Consumers More Selective. The trade-off results also indicate that consumers become more selective following the awareness intervention. For example, for Recycled Content (Figure 5, panel (a3)), the treatment increases the expected no-choice probability at low thresholds (by up to 9.6%) and decreases it at high thresholds (by up to 15.4%) compared to the status quo. A similar pattern is seen for Durability (panel (b3)), where the treatment raises the upper limit of the no-choice probability (by up to 9.6%) and lowers the lower limit (by up to 13.5%). These shifts reflect two opposing dynamics. The increase in no-choice probability at low thresholds stems from the costly trade-offs required to meet increased preferences for Recycled Content—typically achieved by reducing Durability or downgrading the Bill of Materials. Conversely, the decrease in no-choice probability at high thresholds is driven by the overall increase in utility for environmental attributes following the treatment, compounded by the higher no-choice utility induced by increased awareness.

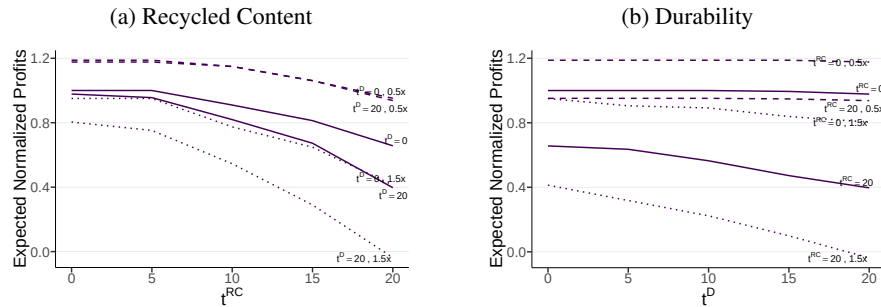
Retailers Substitute Bill of Materials to Offset Sustainability Costs. Finally, we observe a compositional shift in the Bill of Materials from natural to synthetic fibres as environmental improvement constraint thresholds increase. This appears to be a cost-saving strategy adopted by retailers to offset the higher costs of meeting sustainability targets, given limited consumer WTP. The sudden increase in the lower limit of the no-choice probability frontier for Recycled Content, and in both limits for Durability (Figure 5 panels (a3) and (b3) respectively) corresponds with this material shift. Since synthetic fibres are perceived as lower-value by consumers, their inclusion reduces product utility and increases the likelihood of opting out.

5.4. Robustness Analysis on Cost Multipliers

Conversations with industry experts and key opinion leaders indicate substantial variability in costs for each of the three focal environmental attributes. The results presented in previous sections rely on best estimates, but variations could arise due to factors such as provenance, technology (e.g., mechanically-recycled fibres versus chemically-recycled ones), techniques, or the machinery used in garment production. To ensure the robustness of our results, we examine reductions and increases in costs, providing upper and lower limits on business performance and allowing us to explore how product lines are impacted by cost variability.

We focus our discussion on the profitability-environmental impact trade-off frontier for Recycled Content (Figure 6) under various cost scenarios (0.5x, 1x, and 1.5x) in the status quo scenario ($ed^* = 0$). Appendix EC.9 contains the remaining results for Recycled Content and those for Durability.

Figure 6 Trade-Off Frontiers with Modified Costs–Profits ($ed^* = 0$)



Note. (1) Profits: Profits are normalised by the expected profit at $t^{RC} = t^D = t^{CD} = 0$ and $ed^* = 0$. (2) Gross Margin: Gross margin for each optimal product version is computed as $(price - cost)/price$. (3) No-Choice Probability. In all panels, the bold lines correspond to results for the baseline model interactions presented in Figure 5. The dashed lines correspond to results when all costs are halved (0.5x multiplier applied to all costs). The dotted lines correspond to results when all costs are increased by 50% (1.5x multiplier applied to all costs).

Results show a flatter trade-off frontier at lower costs. Results for Durability, and for the increased awareness scenario, are consistent. Moreover, Figure 6 indicates that the 1.5x cost multiplier results are closer to our baseline than the 0.5x cost multiplier results, proving our results to be robust. In fact, outcomes could potentially prove better than reported since we consider our baseline cost estimates to be conservative.

6. Discussion

This paper leverages an experimental and data-driven optimisation approach to study drivers of transitioning toward less environmentally impactful production practices in the fashion industry. We perform an incentive-aligned choice-based conjoint experiment to characterise preferences and willingness-to-pay (WTP) for environmental attributes, including a manipulation aimed at increasing respondent awareness of environmental attributes in apparel. The resulting data inform a joint product line selection-pricing optimisation and counterfactual simulation comparing two scenarios: status quo versus increased consumer awareness. Our findings offer valuable insights as to aligning consumer preferences, retailer trade-offs, and regulatory objectives. This section summarises the key findings and discusses the numerous managerial insights and policy implications related to our study's three main research questions.

Consumer Preferences and Willingness-to-Pay (RQ1). Our first research question seeks to understand preferences and WTP for environmental attributes, relative to base attributes and to each other, under the status quo.

Our study reveals that, despite consumers showing relatively low sensitivity to environmental attributes and displaying inconsistent preferences between direct elicitation and choice experiment results, there remains a clear interest and willingness to pay for sustainability in fast fashion products. We find Price to be the main driver of choice, the second most important being Bill of Materials. Preferences across environmental attributes are similar, with Recycled Content emerging as the most important, followed by Durability and Circularity Design. However, the hierarchy revealed in the choice experiment contrasts with the results from the stated ranking question, highlighting a key inconsistency between consumers' stated beliefs and their behaviour in our incentive-aligned experiment. One might expect Durability to offer more value to consumers, as more durable garments can be worn longer. However, our findings suggest that in the context of fast fashion, Recycled Content is the most preferred attribute and provides the highest utility. This may be due to the immediate reward it offers to consumers as well as the widespread understanding of recycling as a green practice. In contrast, the lower valuation of Durability may reflect temporal discounting, particularly salient in fast fashion, where immediate gratification often outweighs consideration of long-term benefits. This key observation aligns with the findings of [Sun et al. \(2021\)](#), who also identified significant temporal discounting in fashion purchase decisions.

Unlike prior studies that treat sustainability as a binary feature, our multi-level attribute approach reveals that consumer utility behaves like a step function with a sharp increase when moving from zero to the first level of an environmental attribute, and minimal gains beyond that. Our WTP results refine this view, showing a step-to-linearly-increasing pattern, particularly for Recycled Content and Circularity Design. This insight is especially relevant for retailers, as costs typically rise with higher attribute levels. For a profit-maximizing strategy, it may be optimal to consistently offer at least the first level of each environmental attribute, with higher levels considered where the additional cost is justified. We further elaborate on this point by examining misalignment in regulatory incentives below.

Impact of Environmental Constraints on Profitability (RQ2). Our second research question aims to understand how retailer profitability is shaped by improvement constraints across the three environmental attributes. As before, we focus on the status quo where no effort is made to raise consumer awareness.

Our results show that when environmental improvement constraints are imposed sequentially, retailers can enhance their product line assortment without compromising profitability, up to a point. At lower improvement thresholds, they efficiently balance product attributes to maintain high total utility. However, beyond certain thresholds ($t^{RC} \approx 5$, $t^D \approx 15$), the declining utility-to-cost ratio—particularly for Recycled Content—forces trade-offs that negatively affect both profitability and consumer satisfaction.

The interplay between attribute-specific utility-to-cost dynamics and cost structure differences leads to an important insight: the overall impact of environmental constraints on profitability highly depends on which

attribute is constrained. Additionally, it hints at the potential impact of the order in which constraints are introduced. For instance, a retailer initially facing a constraint on Durability—with no other restrictions—may find compliance relatively manageable without sacrificing profitability. However, if a constraint on Recycled Content is introduced later, the additional impact on business performance could be substantial. Conversely, if the first constraint targets Recycled Content, the initial adaptation may be more challenging, but once absorbed, adding a constraint on Durability may have only a marginal effect. This result holds important implications for policymakers, suggesting that the sequencing of regulatory interventions can significantly influence the burden placed on businesses and should be carefully considered in the design of fair and effective sustainability policies.

Impact of Increasing Consumer Awareness (RQ3). Our final research examines the impact of a short educational video (the treatment) on consumer preferences, WTP, and retailer performance.

Impact on consumer preferences and WTP. While raising awareness through a brief intervention proves challenging, consistent with prior findings (e.g., McKie et al. 2023; Steele et al. 2024), we do observe meaningful effects. Notably, the treatment leads to a significant reduction in the importance of the Bill of Materials (BoM), one of the main attributes consumers have relied on for product purchases. This suggests that, following the awareness intervention, consumers' judgments become less anchored in the material composition itself and more influenced by broader considerations, specifically the material's provenance (recycled content), durability, and end-of-life impact (circularity). Though difficult to shift consumer behavior directly, increased awareness is essential for driving participation in circularity initiatives.

We find that greater consumer awareness flattens the trade-off between profitability and environmental impact, as utility and WTP for environmental attributes improve. However, this shift creates a strategic compromise for retailers between Recycled Content, Durability, and BoM. Since Recycled Content remains the most valued attribute, retailers tend to prioritise it, often at the expense of Durability or BoM. This trade-off may incentivise cost-saving choices such as switching from natural to synthetic fibers. This highlights the complexity of policymaking, as regulatory policies must account for shifts in consumer preferences, and consequent retailer reactions under increased awareness treatments.

Impact on Retailer Strategy and Consumer Choice Behavior. The treatment also increases the mean average partworth utility for the no-choice alternative by about 2% (up to 10% in our counterfactual simulation), implying that consumers become more critical of available products. Said differently, the gap between a baseline product and opting out narrows, suggesting that heightened awareness can increase scrutiny and reduce tolerance for high-impact options. However, we also observe that as environmental improvement thresholds rise, the likelihood of purchase increases, potentially stimulating demand for lower-impact products. This paradox raises the concern that increased purchases of “greener” products could still lead to

higher overall consumption, potentially undermining sustainability goals. From a managerial perspective, this presents an encouraging message: engaging consumers through awareness-building can enhance brand image and competitiveness, even if behavioral shifts are gradual.

Misalignment of incentives. An important consideration for regulators is ensuring that policies and incentives are aligned with both the intended sustainability goals and the realities of current market dynamics. In this regard, our study offers three important insights.

First, as discussed above, it may be more effective to implement minimal environmental targets sequentially rather than simultaneously, and gradually rather than abruptly. A phased approach allows retailers to adapt existing product lines and pricing strategies with minimal disruption, while enabling more effective prioritization across environmental dimensions. This, in turn, enhances their ability and agility to meet sustainability goals and reconfigure supply chains accordingly. Moreover, such a progressive roll out offers consumers the opportunity to gradually familiarize themselves with the various environmental attributes of fashion products, fostering more informed and sustainable choices over time.

Second, our results show that optimal assortments are always a mix of high- and low-impact products. This product line strategy is optimal as it allows to meet the tastes of a varied consumer population. For retailers, it is also optimal as it ensures sell-through of products, competitiveness, and can even be used as a way to finance its sustainable transition as high-impact products are likely to sell more, at a higher margin. However, the EU Strategy is currently written in a way that implies *all* products would need a determined minimum level for each of the environmental sustainability dimensions (e.g., recycled content and durability and circularity design). This inflexible approach may prove difficult for brands and retailers to implement, potentially jeopardizing the successful execution of the Strategy.

Finally, we highlight the misalignment between financial incentives and actual market dynamics, using the recent French market as an illustrative example. Consider a retailer aiming to incorporate 30% recycled content into its sweatshirt production. Under current French legislation (developed under the EU Strategy), the retailer would receive an eco-modulation bonus of £0.08/unit, but implementing this change would cost an additional £4.83 per unit. With a consumer willingness-to-pay (WTP) of £11.67, the net added value is substantial at £6.92/unit, suggesting that despite modest subsidies, the market signal from consumer demand is strong enough to motivate adoption. This contrast becomes even more pronounced when viewed alongside recent regulatory developments where the French government has unanimously approved a law⁹ that introduces a €5/unit penalty on fast fashion products (up to €10/unit by 2030). In our example, such a penalty would reduce the net added value from transitioning to recycled content to nearly zero—dramatically weakening the financial rationale for sustainability investments under current incentive structures. This comparison reveals a key tension in policymaking. While eco-modulation bonuses are modest

⁹[Proposition de loi visant à réduire l'impact environnemental de l'industrie textile](#) voted into law on June 10, 2025.

and intended to support responsible production, the newly introduced penalties appear more punitive than transformative. They place a disproportionate financial burden on high-volume, low-cost producers, potentially discouraging even genuine efforts to improve. Without a corresponding increase in positive incentives, such policies risk closing the door on feasible sustainability transition pathways for fast fashion retailers.

Our findings therefore underscore the need for coordinated policy design that aligns with market behavior and consumer willingness to pay. Regulators appear to assume that demand-side enthusiasm will finance most of the transition. As retailers often note, stated consumer interest does not always translate into real-world purchasing behavior. To ensure that sustainability efforts are both impactful and achievable, it is essential to foster an open dialogue among policymakers, retailers, and consumer advocacy groups. Only through such collaboration can we manage the complex trade-offs across environmental dimensions and build fair, effective policies that generate lasting cultural and behavioral change.

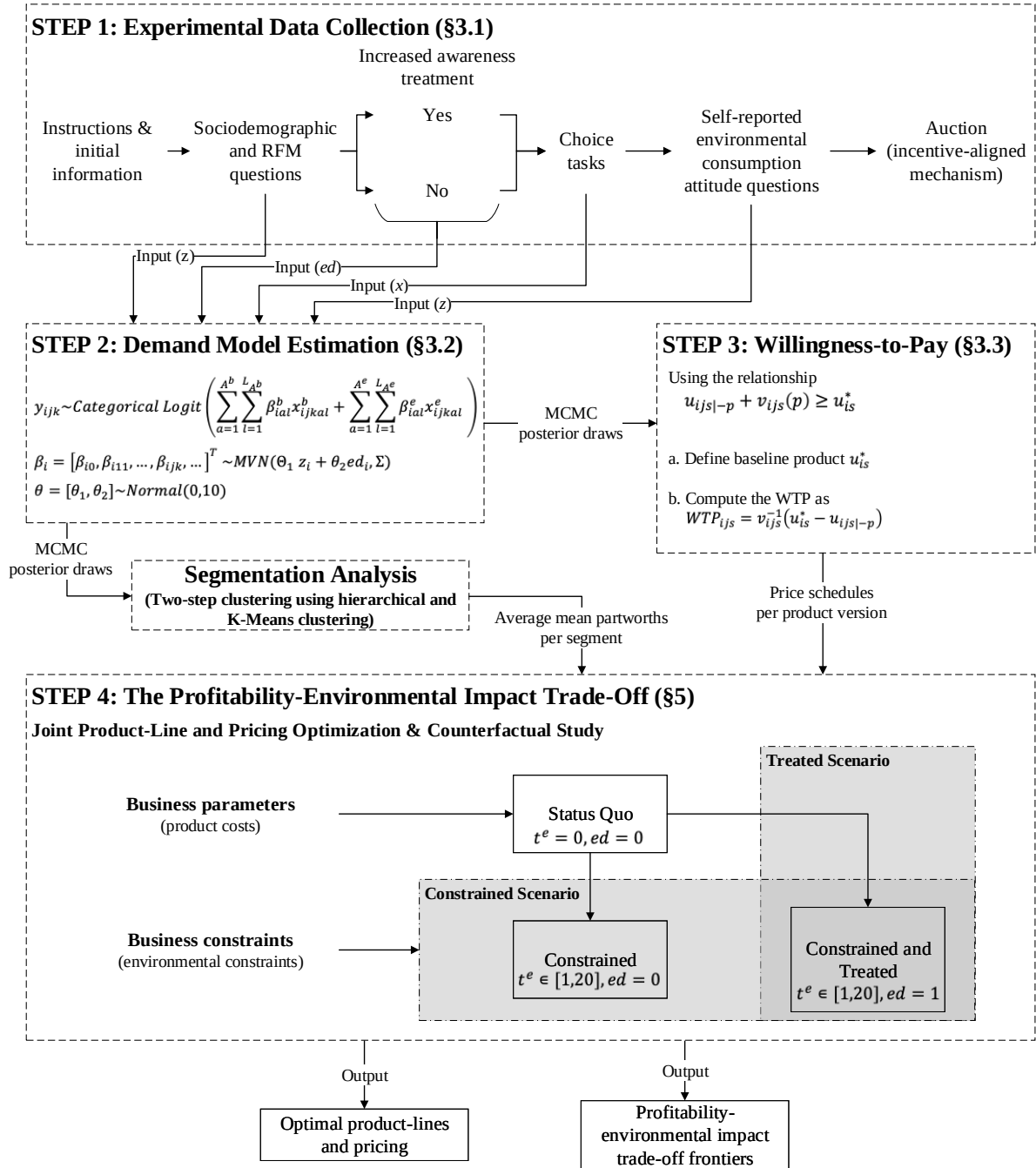
Limitations and Future Research. One of the key findings of this study is that showing respondents a short educational video appears to shift preferences of consumers for environmental attributes, thus easing the profitability-environmental impact trade-off for retailers. However, this simple intervention may not impart long-lasting changes in consumer behaviour ([Osbaldiston and Schott 2012](#)). It would be valuable to measure how much of the information respondents retain after the treatment, and how much of it influences subsequent purchasing decisions. This could transform the treatment into lasting education and literacy. To test this hypothesis, a follow-up choice experiment could be conducted with the same respondents several months after the initial choice experiment. Additional research is required to better understand the broader impact of increased awareness and the various ways in which this can be achieved. Exploring different consumer awareness/education mechanisms—such as educational videos, marketing campaigns, sustainability icons, etc.—offers promising paths for future research.

Another important observation from this study is the high variability and uncertainty in the cost structure for different environmental attributes, highlighting the need for deeper sensitivity analysis to assess how costs impact willingness-to-pay and business performance. Additionally, our choice-based conjoint experiment may have underestimated respondent price sensitivity, potentially skewing certain insights presented here. Fusing experimental data with observational data, such as scanner or clickstream data, could provide a more accurate estimate of consumer price sensitivity.

Finally, while this study assumes no correlation among attributes, this may be an oversimplification since recycled fibers, for instance, may be less durable. Extending these findings to detect correlations among the attributes represents another important direction for future research.

Appendix

Figure A1 Schematic Overview of the Integrated Framework



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Electronic Companion to “Green Synergy: A Three-Party Fashion Retail System Balancing Profit Maximization and Environmental Impact Minimization”

EC.1. Choice-Based Conjoint (CBC) Experiment

Figure EC.1 shows the example sweatshirt shown to respondents at the beginning of the choice experiment. The sweatshirt category satisfies the following criteria, which facilitates data collection and applicability of our study: (a) it is usually offered at most fashion retailers, and usually constitutes one of the largest and most active categories, (b) it is sold at a large range of prices, yet has a price point affordable to every consumer (i.e., everyone can purchase and has a sweatshirt), (c) it is easy to describe in words even if no photo is provided, (d) it is purchased by all genders.

Figure EC.1 Example Sweatshirt



EC.1.1. Choice Set Design

We design two separate sets of choice tasks: an estimation set and a validation set. Each choice task consists of three product alternatives and a no-choice option. The estimation set includes 15 choice tasks, and is used to estimate individual and aggregate level preferences. In addition, respondents complete five fixed validation tasks, which are identical across all participants and serve to assess model fit. In total, each respondent completes 20 choice tasks, and the entire survey was estimated to take approximately 15 minutes to complete.

We design the experiment in a way that alternates between estimation and validation choice sets. This prevents collecting data for the validation sets only at the end of the experiment, when respondents may be more fatigued or distracted. We enforce that every fourth choice set is a validation set, and vary the sequence of validation sets presented to each respondent according to a Latin square design.

Our design of 6 attributes with 5 levels each put us on the higher end of the number of choice tasks to use to collect enough information for estimation and inference. [Johnson and Orme \(1996\)](#) mention that 20 choice tasks is acceptable, and suggest that such a number of choice tasks will not lead to lower data quality. In our case, having more tasks gives a chance to respondents to learn better the attributes, without compromising response quality.

Estimation Choice Sets. We allow for full heterogeneity across respondents by following the method proposed by [Liu and Tang \(2015\)](#), resulting in hyper-individualized (i.e., each respondent will see a unique set of estimation choice tasks) estimation choice set designs with minimal overlap, level balance, and orthogonality. To facilitate comparison of results between experimental treatment condition, we show the same sets for each condition (e.g., respondent 1 of the experimental treatment arm and respondent 1 of the control group will see the exact same sets). Therefore, the heterogeneous design is done for 250 respondents. The method used requires we specify utilities of each level. The attribute levels specified are based on the effects-type coding. We assume a prior specification of mean average partworth utilities (μ in [Liu and Tang \(2015\)](#)) that is linearly decreasing for Bill of Materials and Price, linearly increasing for Recycled Content and Durability, and we assume invariant preferences across the levels for Type and Circularity Design. Our results (see Figure 2) confirm these assumptions. We also assume the variance to be $\sigma = 0.25$ (minimises dominance and overlap). We obtain a design with roughly equal level balance per attribute (i.e., equal frequency of levels per attribute). In about 10% of choice sets (across respondents) have a dominated alternative (i.e., an alternative that attains the same or a less attractive level across all attributes). Finally, due to how the method used distributes choice sets across respondents, there is severe overlap between the first tens of respondents. To counter this overlap, we randomize the respondents before uploading the choice sets to Sawtooth.

Validation Choice Sets. All respondents are shown exactly the same validation set such that we can consistently evaluate the validity of the model. We define the validation sets by following the method proposed by [Gallino et al. \(2019\)](#). We first create 999 choice sets in Sawtooth (the maximum possible number allowed by the software). We then remove any set that appears in the estimation sets, have repeated attribute levels within one choice set, have a dominated alternative, and have extreme expected choice shares (choice probability ≤ 0.1 , or ≥ 0.6).¹ The candidate sets left are examined for face validity, and ensure we obtain 15 levels across all validation sets (5 validation sets of 3 alternatives), meaning each level occurs 3 times on average in validation sets. Finally, we vary the sequence of validation sets presented to each respondent according to a Latin square design.

EC.1.2. Pricing Study

Table [EC.1](#) below presents the range of prices (in GBP) offered at different leading fast fashion retailers in the UK. Pricing information was collected on March 15, 2023.

We observed a price range of £10-50. For the conjoint experiment, we increase the range to £10-70 to allow for a price premium on levels that are not yet available. We confirm the upper bound on our price

¹Our upper bound is lower than that used by [Gallino et al. \(2019\)](#) since we have in our design a no-choice option.

Table EC.1 Fast Fashion Brands Studied and Price Ranges

	Women		Men		Notes
	low	high	low	high	
H&M	10	45	10	50	Excludes ARKET brand
ZARA	10	50	20	60	Excludes ADERxZARA collaboration
Boohoo	10	50	10	50	
Uniqlo	20	30	20	40	
SHEIN	10	40	15	50	

range by sampling conscious fashion brands and to find that prices are in the range of £50-80. The price range is thus defined in a way to mitigate extreme response (Gensler et al. 2012) as it is large enough such that it overlaps with respondents' WTP range, yet small enough that it is not out of range of respondents' expectations. The intermediate price levels are determined by equally splitting the range obtained such that each level is equidistant.

EC.1.3. Ranking Results

Table EC.2 presents results from the ranking question.

Table EC.2 SR-ECA–Summary Statistics

	Treatment Group (N=231)	Control Group (N=258)	Overall (N=489)
Most relevant feature–Rank 1			
Recycled Content	68 (29.4%)	49 (19.0%)	117 (23.9%)
Durability	131 (56.7%)	179 (69.4%)	310 (63.4%)
Circularity Design	32 (13.9%)	30 (11.6%)	62 (12.7%)
Most relevant feature–Rank 2			
Recycled Content	90 (39.0%)	116 (45.0%)	206 (42.1%)
Durability	59 (25.5%)	57 (22.1%)	116 (23.7%)
Circularity Design	82 (35.5%)	85 (32.9%)	167 (34.2%)
Most relevant feature–Rank 3			
Recycled Content	73 (31.6%)	93 (36.0%)	166 (33.9%)
Durability	41 (17.7%)	22 (8.5%)	63 (12.9%)
Circularity Design	117 (50.6%)	143 (55.4%)	260 (53.2%)

EC.2. Hierarchical Bayes Estimation Model

We present in this appendix how we processed the data collected during our experiment (including any data manipulation (e.g., aggregation) and Confirmatory Factor Analysis), and additional estimation results.

EC.2.1. Data Preprocessing

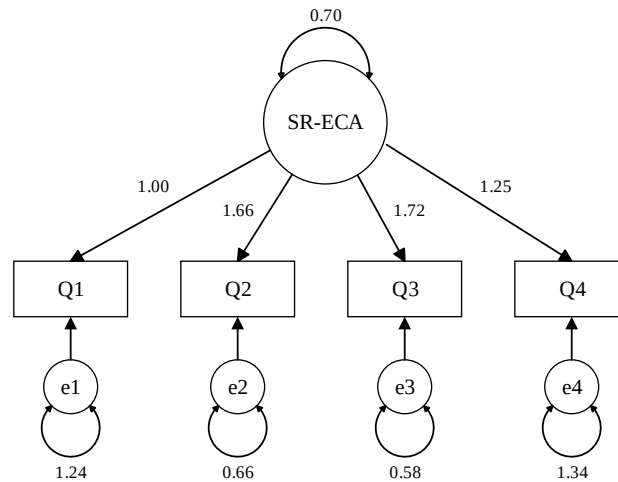
Some of the data collected needed manipulations prior to using it for the estimation of the HB-MNL model. We describe these manipulations here.

We group Gender under three subcategories: Men, Women, and Other. We group Non-Binary and Rather Not Say into the Other category as there were not enough occurrences of each in our dataset. For an identical reason, we group education levels into three subcategories: University Degree, Technical community college or High School diploma A levels, and Other. We group Doctorate degree (PhD/other), Graduate degree (MA/MSc/MPhil/other) and Undergraduate degree (BA/BSc/other) under University Degree. We group Technical/community college and High school diploma/A-levels under Technical community college or High School diploma A levels. Finally, we group Secondary education (e.g. GED/GCSE), No formal qualifications and Don't know / not applicable under Other.

We treat the variables of the SR-RFM questions (three variables with 5 or 8 levels) as continuous from 1 to the maximal level, and include them separately in our model. Similarly, we treat the variables of the SR-ECA questions (four variables with 7 levels) as continuous, and use the numeric values in the Confirmatory Factor Analysis.

EC.2.1.1. Confirmatory Factor Analysis. The latent construct we develop, based on four questions we ask respondent after the choice tasks, is a reflective construct, meaning answers to the questions represent a certain underlying attitude. It is this construct that we include in our estimation model. We use Confirmatory Factor Analysis (CFA) to test the causal SR-ECA model, and to measure, for each respondent, the value of this SR-ECA latent construct (i.e., CFA serves as a dimensionality-reduction method to distill the answers of four questions into one continuous variable). CFA results are summarised in Figure EC.2. Based on the coefficient estimates obtained from the CFA. Our results show good fit of the model.

Figure EC.2 CFA Results Diagram



Note. All results are highly significant (p -value = 0). CFI = 0.995, TLI = 0.984, RMSEA = 0.068, p -value H_0 (RMSEA $\leq$ 0.050) = 0.234, p -value H_0 (RMSEA $\geq$ 0.080) = 0.438.

EC.3. Estimation Results

Table EC.3 presents fit metrics of our model specification. We report the hit rate and hit probability for in-sample and out-of-sample fit, in addition to the mean absolute error (MAE). Table EC.4 presents mean average posterior partworth utility estimates of the preference shifters θ_2 .

Table EC.3 Fit Metrics	
Metric	
Estimation sample	
Hit rate	0.886
Hit probability	0.818
Holdout sample	
Hit rate	0.595
Hit probability	0.763
Mean absolute error (MAE)	13.377 %

Note. Hit rates and probabilities are individual-level fit metrics, and correspond to the number of times a preferred alternative is correctly estimated, and the estimated choice probability of the preferred alternative respectively. Hit rates and hit probabilities reported in the table below are averaged across respondents and across posterior draws. The MAE is an aggregate-level measure of fit and is calculated as the mean absolute error between actual choice frequencies and estimated frequencies across holdout sets.

Table EC.4 Posterior Partworth Utility Estimates—Preference Shifters (θ_2)		
	Mean	St. Dev.
$\theta_2 - \text{BoM:75\%polyester,25\%cotton}$	0.174	0.262
$\theta_2 - \text{BoM:50\%polyester,50\%cotton}$	-0.391	0.262
$\theta_2 - \text{BoM:25\%polyester,75\%cotton}$	-0.461	0.278
$\theta_2 - \text{BoM:100\%cotton}$	-0.427	0.324
$\theta_2 - \text{Type:Lounge}$	0.477	0.263
$\theta_2 - \text{Type:Casual}$	0.047	0.240
$\theta_2 - \text{Type:Active}$	-0.260	0.253
$\theta_2 - \text{Type:Essential}$	-0.266	0.225
$\theta_2 - \text{RC:15\%recycled.content}$	-0.160	0.260
$\theta_2 - \text{RC:30\%recycled.content}$	0.555	0.233
$\theta_2 - \text{RC:45\%recycled.content}$	0.416	0.243
$\theta_2 - \text{RC:60\%recycled.content}$	0.805	0.231
$\theta_2 - \text{D:10\%increase}$	0.006	0.256
$\theta_2 - \text{D:20\%increase}$	0.176	0.235
$\theta_2 - \text{D:30\%increase}$	0.442	0.246
$\theta_2 - \text{D:40\%increase}$	0.190	0.237
$\theta_2 - \text{CD:Designed.for.repairs}$	0.047	0.248
$\theta_2 - \text{CD:Designed.for.redistribution}$	0.467	0.245
$\theta_2 - \text{CD:Designed.for.upcycling}$	0.418	0.229
$\theta_2 - \text{CD:Designed.for.recycling}$	0.390	0.216
$\theta_2 - \text{Price:55}$	0.483	0.298
$\theta_2 - \text{Price:40}$	0.155	0.272
$\theta_2 - \text{Price:25}$	0.107	0.325
$\theta_2 - \text{Price:10}$	-0.095	0.390
$\theta_2 - \text{No-Choice}$	0.542	0.463

Note. Values reported correspond to the posterior mean (across posterior draws) average (across respondents) partworth utility estimates. Values in bold exclude zero in the 95% credible interval of the posterior draws.

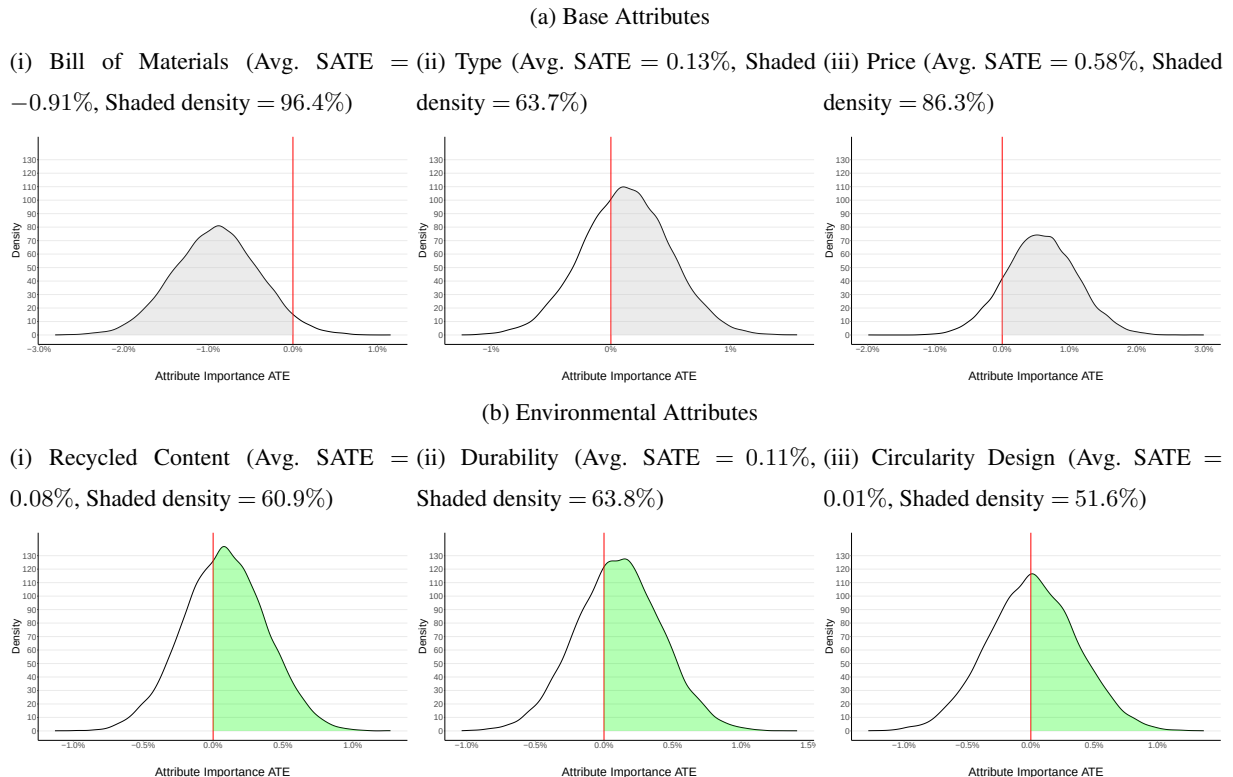
EC.4. Average Treatment Effect.

To measure the causal effect of receiving the treatment, assuming perfect randomization, we compute the Sample Average Treatment Effect (SATE) as:

$$SATE_s = \mathbb{E}[g(\beta_{is}|ed_i = 1) - g(\beta_{is}|ed_i = 0)] = \frac{1}{N_1} \sum_{i_1 \in \{i|ed_i=1\}} g(\beta_{i_1s}) - \frac{1}{N_0} \sum_{i_0 \in \{i|ed_i=0\}} g(\beta_{i_0s}) \quad (\text{EC.1})$$

where s refers to a posterior draw, and $g(\cdot)$ is a nonlinear function that transforms partworth utilities into attribute importance (using eq. (4)) or WTP (following Algorithm 1), $ed_i = 0$ corresponds to respondents in the control group, $ed_i = 1$ corresponds to respondents in the treatment group, N_0 and N_1 correspond to the number of respondents in each respective group, and β_{i_0s} and β_{i_1s} denote the set of posterior partworth utilities for respondents per respective group. Figure EC.3 presents the posterior distributions of the difference in average attribute importance per attribute (i.e., the posterior distribution of the difference between the light and dark grey bars per attribute in Figure 4).

Figure EC.3 Posterior Attribute Importance SATE Distribution



Note. The SATE on attribute importance is computed by applying eq. (EC.1), where $g(\cdot)$ here represents the attribute importance calculation. The average SATE is computed across posterior draws.

Figure EC.3 panel (a-i) shows a difference of 0.91% in attribute importance between treatment groups corresponding to a significant decrease of 4.3 percentage points in average attribute importance for BoM

with treatment, for which 96.4% of the distribution density below 0 (i.e., for 96.4% of the posterior distribution (across respondents and draws), showing the video decreased the average attribute importance of BoM). Moreover, Figure EC.3 panel (b) shows gains in attribute importance for all environmental attributes, the largest being for Durability (panel (b-ii), 1.03 percentage points increase) with 63.8% of the distribution density above 0, while moderate for Recycled Content (panel (b-i), 0.65 percentage points increase) with 60.9% of the distribution density above 0. There is almost no difference for Circularity Design (panel (b-iii), 0.08 percentage points increase) with 51.6% of the distribution density above 0.

EC.5. Measuring Willingness-to-Pay and Algorithms

This appendix presents a pseudo-code for the WTP calculation as well as a short discussion on the choice of appropriate baseline product u_i^* for each of the calculations carried in our study.

Algorithm 1 below presents a pseudo-code for the computation of the WTP. The codes (in R) for implementation can be found on the project's OSF page. Recall that the WTP is calculated using equation

$$u_{ijs|-p} + \nu_{ijs}(p) \geq u_{is}^* \implies WTP_{ijs} = \nu_{ijs}^{-1}(u_{is}^* - u_{ijs|-p})$$

where $u_{ijs|-p}$ corresponds to the total utility, excluding the utility of price, respondent i attaches to product j for posterior draw s , $\nu_{ijs}(p)$ corresponds to the utility of price p respondent i attaches to product j for posterior draw s , and u_{is}^* corresponds to the total utility of a baseline for respondent i for posterior draw s against which we compare product j .

The computation of WTP using this approach relies on comparing consumer utilities for two competing products—a baseline product, and a focal product. The choice of an adequate baseline is therefore crucial for the accuracy of the computation and interpretation. In their study, Kohli and Mahajan (1991) define the baseline product as the most preferred product in the consumer's evoked set (i.e. the most preferred product of all products currently available to a consumer in a market). For each of the three WTP calculations in this study, we carefully define the baseline alternative against which the WTP is computed. We discuss the choice of each baseline below.

The first WTP calculation we perform is used to measure the main effects incremental WTP for each level of each environmental attribute (i.e., the incremental monetary amount a consumer would be willing to pay between a product with no environmental attributes, to one with a given level of a given environmental attribute). Contrary to Kohli and Mahajan (1991), it is impossible for us to define a market standard for the sweatshirt category. In addition, since we consider consumer heterogeneity in our study, each respondent may not have the same preferred product (Breidert et al. 2006). Consequently, for this calculation, the baseline alternative is defined as the most preferred base product (i.e., the combination of maximal partworth

Algorithm 1 WTP Calculation Algorithm

```

1: Using the estimated partworth utilities from CBC responses.
2: for each treatment group (if applicable) do
3:   for each posterior draw  $s$  do
4:     for each respondent  $i$  do
5:       Define  $u_{is}^*$ .
          • For the incremental WTP calculation,  $u_{is}^* = \max_{l_{BoM}} u_{isl_{BoM}} + \max_{l_{Func}} u_{isl_{Func}} + v_{is}(34.50)$ , where  $l_{BoM}$  and  $l_{Func}$  correspond to levels of the Bill of Material and Functionality respectively, and  $v_{is}(34.50)$  is the utility level for a sweatshirt priced at £34.50 for respondent  $i$  and draw  $s$ .
          • For the auction product WTP calculation,  $u_{is}^* = u_{0is}$ , where  $u_{0is}$  corresponds to the utility of the outside option for respondent  $i$  and draw  $s$ .
          • For the product versions WTP calculation,  $u_{is}^* = u_{0is}$ .
6:       Define  $u_{ijs|-p}$  the total utility of the focal product excluding the utility of price.
          • For the incremental WTP and product versions WTP calculations,  $u_{i|-p} = \max_{l_{BoM}} u_{isl_{BoM}} + \max_{l_{Type}} u_{isl_{Type}} + u_{i|e^l}$ , where  $u_{i|e^l}$  corresponds to the partworth utility of the focal environmental attribute level. This calculation needs to be repeated for each environmental attribute and each level.
          • For the auction product WTP calculation,  $u_{is|-p} = x^{auction} \beta_{is|-p}$ , where  $\beta_{is|-p}$  corresponds to the estimated partworth utilities for all attributes except price for respondent  $i$  and posterior draw  $s$ , and  $x^{auction}$  is a binary vector encoding the levels defining the auction product.
7:       Compute  $WTP_{ijs} = \nu_{ijs}^{-1}(u_{is}^* - u_{ijs|-p})$  by linear interpolation.
          For respondents with  $WTP_{ijs}$  falling outside the range of price levels, assign  $WTP_{ijs} = 9.99$  or  $WTP_{ijs} = 70.01$ .
8:     end for
9:   end for
10: end for

```

of Bill of Materials and Type only) for each respondent for each posterior draw. We argue that such a baseline product over the no-choice option in this case is more suitable because it is the most efficient methodological approach to isolate the WTP for each environmental attribute level. We use an average price of sweatshirt of £34.50² instead of the partworth utility for price to avoid the trivial result that all respondents prefer the lowest (or lower) price level. Using the average price therefore provides us with an adequate reference point to compute the WTP. The result is a sweatshirt product with no environmental attributes, providing us with a satisfactory baseline product.

The second WTP calculation we perform is used to determine the bid price of each respondent for the auction. For this calculation, the baseline alternative is defined as the no-choice alternative.

The final WTP calculation we perform is used to calculate the mean average WTP of each product version by using the posterior draws from our estimation simulation. This allows us to fully leverage the measure preference information to guide the retailer in defining sensible price ladders for each product that includes new features. For this WTP calculation, we use as a baseline the no-choice alternative as a retailer will want to price new products such that consumers would want to purchase them (i.e., compete with not purchasing anything, or purchasing from a competitor).

Finally, for our application, we assume a piecewise linear function for $\nu_{ijs}(p)$, defined by the price partworth utilities estimated, and use linear interpolation to find the WTP. In the event a WTP is estimated outside the range of price levels contained in our experiment (i.e., lower than £10, or greater than £70), we constrain the WTP to £10 or £70.

EC.5.1. Additional WTP Results

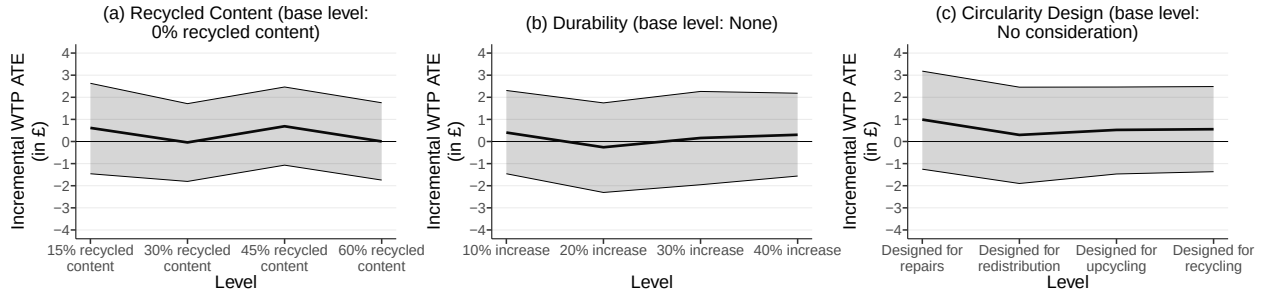
Figure EC.4 summarises the posterior incremental WTP average treatment effect (ATE) distributions for each environmental attribute.

EC.6. MILP Formulation

We present in this appendix the derivation of our proposed MILP reformulation (jPLSP-MILP). Our main goal is to derive a strong reformulation that allows to efficiently deal with the expectation in the objective function, to use commercial solvers, and to have a decent runtime given that we run in total 900 different models. Our proposed reformulation relies on the continuous linear reformulation discussed in [Avadhanula et al. \(2016\)](#) and [Sumida et al. \(2021\)](#), which we extend for a mixture of MNL to capture consumer heterogeneity in preferences.

First assuming that $\mathbb{E}_\beta [D_{qr}(x)] = \frac{u_{qr}}{u_0 + \sum_{q',r'} u_{q'r'}}$ and letting $y_{qr} = \frac{v_{qr}x_{qr}}{v_0 + \sum_{q',r'} v_{q'r'}x_{q'r'}}$ and $y_0 = \frac{v_0}{v_0 + \sum_{q',r'} v_{q'r'}x_{q'r'}}$, where $v_{qr} = \exp(u_{qr})$ and $v_0 = \exp(u_0)$, the continuous reformulation of our jPLSP model 5 is written as:

²From [Statista \(2023\)](#); Women's per-unit price: £35.18 (Europe). Men's per-unit price: £33.85 (Europe). £35 is also roughly the average price we estimate from our pricing study of the major fast fashion brands in the UK (see Appendix EC.1.2).

Figure EC.4 Posterior Incremental Willingness-to-Pay SATE Distribution

Note. The SATE on incremental WTP is computed by applying eq. (EC.1), where $g(\cdot)$ here represents the incremental WTP calculation for each level of each environmental attribute. Bold lines indicate the mean (across posterior draws) SATE per level. Shaded areas indicate the 95% credible interval.

$$(jPLSP - cont.) \quad \max_y \mathbb{E} \left[\sum_q \sum_r (p_{qr} - c_q) y_{qr} \right] \quad (EC.2a)$$

$$s.t. \quad \sum_r \frac{y_{qr}}{v_{qr}} \leq \frac{y_0}{v_0} \quad \forall q \quad (EC.2b)$$

$$\sum_q \sum_r \frac{y_{qr}}{v_{qr}} = 5 \frac{y_0}{v_0} \quad (EC.2c)$$

$$\sum_q \sum_r IMP_q^{RC} \frac{y_{qr}}{v_{qr}} \geq t^{RC} \frac{y_0}{v_0} \quad (EC.2d)$$

$$\sum_q \sum_r IMP_q^D \frac{y_{qr}}{v_{qr}} \geq t^D \frac{y_0}{v_0} \quad (EC.2e)$$

$$\sum_q \sum_r IMP_q^{CD} \frac{y_{qr}}{v_{qr}} \geq t^{CD} \frac{y_0}{v_0} \quad (EC.2f)$$

$$y_{qr} v_0 \leq y_0 v_{qr} \quad \forall q, \forall r \quad (EC.2g)$$

$$y_0 + \sum_q \sum_r y_{qr} = 1 \quad (EC.2h)$$

Given how we define the new decision variable y , jPLSP-cont. provides us directly with choice probabilities for each product version. It is easy to see, by substituting y_{qr} and y_0 in constraints (EC.2b) to (EC.2g), that we recover exactly constraints (5d) to (5f). Constraint (EC.2h) ensures the choice probabilities sum to 1. Sumida et al. (2021) show that if and only if the constraint matrix is totally unimodular, then the continuous reformulation is equivalent to the initial integer program.

However, in our model, the expectation in the objective function of the jPLSP model 5 is written as:

$$\mathbb{E}_\beta [D_{qr}(x)] = \sum_{i=1}^I \left[\frac{1}{S} \sum_{s=1}^S \frac{\exp(u_{iqr,s}) x_{qr}}{\exp(u_{i0,s}) + \sum_{q'} \sum_{r'} \exp(u_{iq'r',s}) x_{q'r'}} \right] \quad (EC.3)$$

where $u_{iqr,s}$ is the utility respondent i attaches to product version q at price r in posterior draw s , and $u_{i0,s}$ the utility of the no-choice alternative for posterior draw s . Eq. EC.3 uses *all* respondents and posterior

draws in the expectation resulting in an untractable number of terms and variables in the objective function (and model). We simplify the objective function through the plug-in approach and post-hoc segmentation, yielding:

$$\tilde{D}_{qr}(x) = \sum_{m=1}^M \pi_m \frac{\exp(u_{qrm})x_{qr}}{\exp(u_{0m}) + \sum_{q'} \sum_{r'} \exp(u_{q'r'm})x_{q'r'}} \quad (\text{EC.4})$$

where u_{qrm} is the utility segment m attaches to product version q at price q , and u_{0m} the utility for the no-choice alternative. Here, the utility for segment m is the average utility across all posterior draws s of respondents in segment m . The result of this simplification is a mixture of MNLs, one per segment. Incorporating this simplified demand specification back into jPLSP-cont. yields:

$$(jPLSP - MILP) \quad \max_{x,y} \sum_m \pi_m \left[\sum_q \sum_r (p_{qr} - c_q) y_{qrm} \right] \quad (\text{EC.5a})$$

$$s.t. \sum_r x_{qr} \leq 1 \quad \forall q \quad (\text{EC.5b})$$

$$\sum_r \frac{y_{qrm}}{v_{qrm}} \leq \frac{y_{0m}}{v_{0m}} \quad \forall q, \forall m \quad (\text{EC.5c})$$

$$\sum_q \sum_r x_{qr} = 5 \quad (\text{EC.5d})$$

$$\sum_q \sum_r \frac{y_{qrm}}{v_{qrm}} = 5 \frac{y_{0m}}{v_{0m}} \quad \forall m \quad (\text{EC.5e})$$

$$\sum_q \sum_r IMP_q^{RC} x_{qr} \geq t^{RC} \quad (\text{EC.5f})$$

$$\sum_q \sum_r IMP_q^D x_{qr} \geq t^D \quad (\text{EC.5g})$$

$$\sum_q \sum_r IMP_q^{CD} x_{qr} \geq t^{CD} \quad (\text{EC.5h})$$

$$y_{0m} + \sum_q \sum_r y_{qrm} = 1 \quad \forall m \quad (\text{EC.5i})$$

$$x_{qr} \geq y_{qrm} \quad \forall q, \forall r, \forall m \quad (\text{EC.5j})$$

$$y_{qrm} v_{0m} \leq y_{0l} v_{qrm} \quad \forall q, \forall r, \forall m \quad (\text{EC.5k})$$

$$x_{qr} \in \{0, 1\} \quad \forall q, \forall r \quad (\text{EC.5l})$$

Since we are mixing between multiple consumer segments, we reintroduce decision variable x to enforce the retailer to choose the same assortment of product versions to show to *all* consumers, rather than one assortment per segment. Forcing constraints (EC.5j) are also added to ensure a version cannot be presented to segment m if it is not globally chosen.

We note that empirically, we find our reformulation is much faster to solve (a few second for most, 5 minutes at most on a MacBook Pro M1 chip) compared to the conic reformulation presented by [Şen et al. \(2018\)](#) (which, after 2 hours, was still at $> 5\%$ MIP gap on a MacBook Pro M1 chip).

EC.7. Defining Product Costs

To be consistent with how we derived WTP in §3.3, we use the average price of a sweatshirt sold in Europe (without any environmental attributes) of £34.50 and assume a 60% gross margin³ to set the cost of a base product at £13.80. The cost of a product version increases by multiplying the base cost by the relevant cost multipliers corresponding to the level of each environmental attribute and Bill of Materials of the product version.

Bill of Materials. We assume polyester fibre to be twice as cheap as cotton, and assume fabric accounts for 60% of the total cost of a product⁴, resulting in a cost multiplier ranging between 0.8 for 100% polyester product versions, to 1.2 for 100% cotton product versions.

Recycled Content. Following multiple conversations with industry experts in fibre-to-fibre recycling, we understand supply is not on a large industrial scale yet.⁵ Consequently, including fibre-to-fibre recycled material is costly. Industry analyses put recycled fibre costs at about 1.2x to 3x the cost of virgin fibres⁶, the substantial variability in costs depending on the fibre blend, provenance (post-industrial or post-consumer), type of recycling (mechanical or chemical), etc. Acknowledging this industry information, we take an average value for the price of recycled content at 2.1x the price of virgin fibres and calculate the cost multipliers for each level of recycled content we study. To these multipliers, we add 10% to account for the potential high “buy-in” price for including this technology in the product line (updating suppliers, adapting supply chain and contracts, etc.). The result is an exponential cost structure that capture the increasing barriers in sourcing a larger amount of recycled fibres when supply is low.

Durability. The durability of fashion products depends on all aspects of the product: from the raw material, to the machinery used to create the garment, to the skill of workers, quality of materials used to put it together, to the dyeing process⁷. Consequently, again, there is substantial variability in costs given the many options a retailer can choose from. For the purpose of our study, we assume that increased durability directly depends on the quality of the fabric, the sturdiness of the garment (i.e., thicker material will last

³As reported recently by [CNBC](#), Zara’s gross margin is 58%. [This source](#) also implies that the gross margin from a sweater (from Zara) is around 60%

⁴See <https://techpacker.com/blog/design/everything-you-need-to-know-about-garment-costing-and-pricing/>

⁵Accelerating Circularity, a leader in fibre-to-fibre recycling, is only [working on small scale batches](#). [Re&Up](#), another leader focusing on cotton and polyester, is also only starting to scale operations.

⁶See for example [this report from RegenFabric](#)

⁷For example, see [WRAP’s clothing longevity protocol](#) for an idea of all dimensions affecting durability.

longer), better stitching and sewing, and more skilled workers, which all translates in more material and more time to make the garment. Consequently, we define a linear relationship for the variation of cost for higher levels of durability. The cost of quality is estimated to range between 20-25% of the cost of a garment⁸, the cost for better quality fibres is estimated to be between 30-50% of the fabric price⁹ (i.e., 18-30% of the total cost), and the cost of improved fabrication ranges between 10-30%. We assume the increase in price is commensurate to the increase in lifespan, and add 5% to capture the buy-in price for the retailer to offer an enhanced durability in its products (e.g., develop and implement enhanced quality control program and various administrative overhead).

Circularity Design. For Designed for repairs, we base our multiplier on a repair scheme of a free lifetime warranty. We assume the cost of repair to be £15¹⁰. We further assume a product would be repaired only once in its lifetime with a probability of 0.1.¹¹ Consequently, on average, it would cost £1.5 to repair a product, which translates to approximately 10% of the product cost.

For Designed for redistribution, we assume retailers would need to incentivise consumers to bring back their old products in order to recirculate them. McKie et al. (2023) find that a £4 (\$5) fee constitutes a moderate fee that has the most impact on participation to a redistribution scheme. We assume 50% of the products would be returned for redistribution and that 90% of the products would be in good enough condition to be redistributed. Therefore, the cost multiplier for Designed for redistribution is 13% of the product cost, inclusive of running costs of a sorting facility.

For Designed for upcycling, we assume there is an already-existing collection and sorting infrastructure in place. In such a case, using and maintaining such an infrastructure bears a marginal per unit cost, and we set the cost multiplier to 5% of the product cost. Note that we focus on post-consumer waste in this study.

For Designed for recycling, we assume that there are already EPR policies and garment collection and recycling infrastructure in place. Then, similar to the electronics industry, the cost incurred by a retailer for each product put on the market is marginal, and we set the base cost multiplier at 5% of the product cost (£0.65).¹² We recognise that the collection, sorting and/or recycling infrastructures in the fashion industry are not yet in place, and will require a substantial investment. However, given the large volume of units produced every year (in the billions for the likes of Zara and H&M), investments amortised per unit sold will still be marginal.

⁸According to [this source](#). Recall that quality and durability are highly correlated, meaning that increases in overall quality lead to increases in durability

⁹According to [this source](#)

¹⁰The French [Écobalyse](#) sets the average price of repair for a pullover at €15.

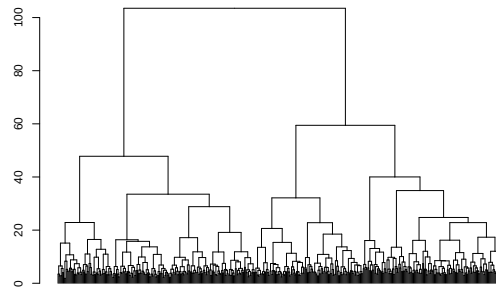
¹¹We consider this probability to be conservative. For example, Patagonia repairs around 50,000 items a year according to [this source](#). Production volume is a datum that manufacturers seldom share.

¹²In the European Union, it is [estimated at €0.12/item](#), while [SistemiQ](#) states an EPR scheme adds about €0.15 to a polyester jumper.

EC.8. Segmentation Results

Figure EC.5 presents the dendrogram of the Hierarchical clustering procedure. From the dendrogram, we find that the optimal number of clusters in our dataset is $M = 2$. However, Figure EC.5 shows those two clusters nicely subdivide into two clusters each, and we decide to use $M = 4$ for our study. Table EC.5 presents descriptives of the four segments we obtain from the two-step segmentation analysis.

Figure EC.5 Hierarchical Clustering Dendrogram–Ward linkage



EC.9. Additional Robustness Analysis Results

Figure EC.6 presents, for Recycled Content and Durability respectively, results for the gross margin and no-choice probability trade-off frontiers omitted in the main body of the paper for different cost scenarios (0.5x, 1x, and 1.5x) and the status quo scenario ($ed^* = 0$) only.

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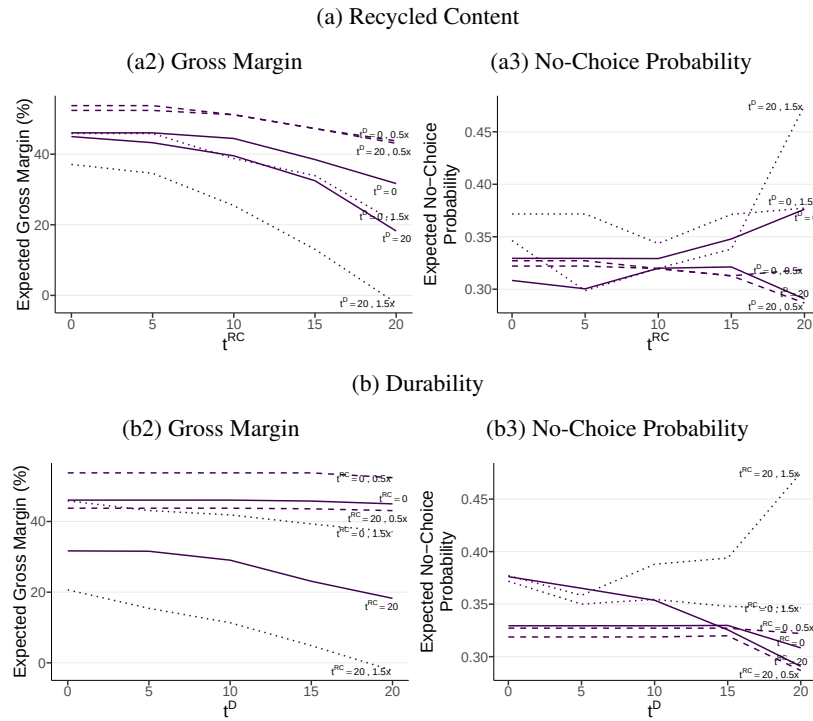
Table EC.5 Descriptives of Segments

	Price-Sensitive	Traditional	Price-Agnostic	Env.-Conscious	Overall
Gender					
Man (including Trans Male/Trans Man)	72 (45.9%)	70 (56.9%)	49 (43.4%)	43 (44.8%)	234 (47.9%)
Non-binary (would like to give more detail)	2 (1.3%)	1 (0.8%)	1 (0.9%)	1 (1.0%)	5 (1.0%)
Rather not say	1 (0.6%)	2 (1.6%)	1 (0.9%)	2 (2.1%)	6 (1.2%)
Woman (including Trans Female/Trans Woman)	82 (52.2%)	50 (40.7%)	62 (54.9%)	50 (52.1%)	244 (49.9%)
Education					
Doctorate degree (PhD/other)	5 (3.2%)	7 (5.7%)	4 (3.5%)	4 (4.2%)	20 (4.1%)
Graduate degree (MA/MSc/MPhil/other)	34 (21.7%)	25 (20.3%)	20 (17.7%)	15 (15.6%)	94 (19.2%)
High school diploma/A-levels	28 (17.8%)	13 (10.6%)	27 (23.9%)	16 (16.7%)	84 (17.2%)
No formal qualifications	1 (0.6%)	1 (0.8%)	0 (0%)	0 (0%)	2 (0.4%)
Secondary education (e.g. GED/GCSE)	19 (12.1%)	11 (8.9%)	12 (10.6%)	7 (7.3%)	49 (10.0%)
Technical/community college	13 (8.3%)	8 (6.5%)	9 (8.0%)	13 (13.5%)	43 (8.8%)
Undergraduate degree (BA/BSc/other)	57 (36.3%)	58 (47.2%)	41 (36.3%)	40 (41.7%)	196 (40.1%)
Don't know / not applicable	0 (0%)	0 (0%)	0 (0%)	1 (1.0%)	1 (0.2%)
SES					
Mean (SD)	5.23 (1.48)	5.23 (1.60)	5.09 (1.47)	5.01 (1.78)	5.15 (1.57)
Median [Min, Max]	5.00 [2.00, 8.00]	5.00 [2.00, 10.0]	5.00 [1.00, 8.00]	5.00 [1.00, 9.00]	5.00 [1.00, 10.0]
Age					
Mean (SD)	33.2 (7.09)	33.0 (7.11)	33.6 (7.33)	33.4 (7.08)	33.3 (7.13)
Median [Min, Max]	34.0 [18.0, 45.0]	33.0 [19.0, 45.0]	35.0 [18.0, 45.0]	33.5 [19.0, 45.0]	34.0 [18.0, 45.0]
RFM-Q1					
Mean (SD)	1.86 (1.03)	2.03 (1.04)	1.82 (0.984)	1.84 (0.988)	1.89 (1.02)
Median [Min, Max]	2.00 [1.00, 5.00]	2.00 [1.00, 5.00]	2.00 [1.00, 5.00]	2.00 [1.00, 5.00]	2.00 [1.00, 5.00]
RFM-Q2					
Mean (SD)	3.62 (1.02)	3.45 (1.00)	3.59 (1.05)	3.46 (1.01)	3.54 (1.02)
Median [Min, Max]	4.00 [1.00, 5.00]	3.00 [1.00, 5.00]	4.00 [2.00, 5.00]	3.00 [1.00, 5.00]	3.00 [1.00, 5.00]
RFM-Q3					
Mean (SD)	3.13 (1.35)	3.27 (1.29)	3.42 (1.43)	3.16 (1.36)	3.24 (1.36)
Median [Min, Max]	3.00 [1.00, 7.00]	3.00 [1.00, 7.00]	3.00 [1.00, 8.00]	3.00 [1.00, 8.00]	3.00 [1.00, 8.00]
SR-ECB-Q1					
Mean (SD)	4.10 (1.34)	4.11 (1.45)	4.24 (1.43)	4.25 (1.38)	4.17 (1.40)
Median [Min, Max]	5.00 [1.00, 7.00]	5.00 [1.00, 7.00]	5.00 [1.00, 7.00]	5.00 [1.00, 7.00]	5.00 [1.00, 7.00]
SR-ECB-Q2					
Mean (SD)	3.51 (1.63)	3.50 (1.64)	3.71 (1.61)	3.66 (1.56)	3.58 (1.61)
Median [Min, Max]	3.00 [1.00, 7.00]	3.00 [1.00, 7.00]	4.00 [1.00, 7.00]	3.00 [1.00, 7.00]	3.00 [1.00, 7.00]
SR-ECB-Q3					
Mean (SD)	3.89 (1.67)	4.00 (1.65)	4.44 (1.60)	4.16 (1.52)	4.10 (1.63)
Median [Min, Max]	4.00 [1.00, 7.00]	4.00 [1.00, 7.00]	5.00 [1.00, 7.00]	4.00 [1.00, 7.00]	4.00 [1.00, 7.00]
SR-ECB-Q4					
Mean (SD)	3.99 (1.52)	3.98 (1.67)	4.36 (1.64)	4.14 (1.37)	4.10 (1.56)
Median [Min, Max]	4.00 [1.00, 7.00]	4.00 [1.00, 7.00]	5.00 [1.00, 7.00]	4.00 [1.00, 7.00]	4.00 [1.00, 7.00]
Condition					
Control	73 (46.5%)	42 (34.1%)	73 (64.6%)	70 (72.9%)	258 (52.8%)
Treatment	84 (53.5%)	81 (65.9%)	40 (35.4%)	26 (27.1%)	231 (47.2%)

Statista (2023) Consumer marketing insights—apparel—jerseys, sweatshirts & pullovers. "<https://www-statista-com/outlook/cmo/apparel/women-s-apparel/jerseys-sweatshirts-pullovers/europe#price>" and "<https://www-statista-com/outlook/cmo/apparel/men-s-apparel/jerseys-sweatshirts-pullovers/europe#price>", (accessed: 05-10-2023).

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Figure EC.6 Trade-Off Frontiers with Modified Costs

Note. (1) Profits: Profits are normalised by the expected profit at $t^{RC} = t^D = t^{CD} = 0$ and $ed^* = 0$. (2) Gross Margin: Gross margin for each optimal product version is computed as $(price - cost)/price$. (3) No-Choice Probability. In all panels, the bold lines correspond to results for the baseline model interactions presented in Figure 5. The dashed lines correspond to results when all costs are halved (0.5x multiplier applied to all costs). The dotted lines correspond to results when all costs are increased by 50% (1.5x multiplier applied to all costs).